

FERRERS-CELL FORMULAS FOR CHAPOTON’S Q-ZETA NUMERATORS: BALLOT WORDS, INTERVAL STRUCTURE, AND TRAPEZOIDAL SPECTRA

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ABSTRACT. Chapoton’s q-Zeta invariant attaches a rank-sensitive bivariate numerator to a finite graded poset. We prove a universality theorem for the augmented Ferrers-cell lattices $F_{\mathbf{b}} = \{(i, c) : i \leq c \leq b_i\}$, where $b_1 \geq \dots \geq b_r \geq r$. For every such boundary, and for every interval, the numerator is an explicit positive descent enumerator over boundary-admissible ballot words. The proof is constructive: a signed EL-labelling identifies descents with the initial positions of left runs, and the full multivariate descent-set polynomial is recovered by a finite transfer-matrix recursion.

For the trapezoidal family $T_{r,s}$, we obtain a Gaussian-binomial formula for every interval and every t -slice. At $q = 1$, the global profile is a Jacobi-polynomial transform; its zeros are simple and negative, strictly interlace along every fixed-offset rank sequence, and have an explicit arcsine push-forward limit. The associated bivariate kernel and its constant-term extraction give algebraic generating functions and the common fixed-offset growth rate $(1 + \sqrt{t})^2$ for $t \geq 0$.

The balanced boundary recovers the standard positive-root posets of types B_r and C_r , graded by root height minus one and normalized with Chapoton’s fixed denominator. Thus, for every Lie rank, we obtain the reversed-ballot formula, coefficientwise q-positivity, the specialization $[t^k]\mathbb{H}_{P_r, \text{rk}}(1, t) = \binom{r-1}{k}^2$, and sharp slice degrees with unique leading monomials. Type A_r is the rectangular specialization. Together these results identify a concrete Ferrers-cell mechanism behind the q-Zeta positivity phenomenon and expose its interval, algorithmic, and spectral consequences.

The main results of this paper were obtained through a generative-AI workflow using OpenAI GPT-5.6 Sol, Anthropic Claude Fable 5, and Grok 4.6. Further details appear in the disclosure at the end of the paper.

1. INTRODUCTION

Positive-root posets connect the combinatorics of finite posets with the structure of crystallographic root systems. Their antichains and order ideals occur naturally in the theory of generalized Catalan numbers and Coxeter arrangements; see, for example, [1, 11]. This makes root posets a natural test class for rank-sensitive refinements of classical order invariants.

Chapoton’s q-Zeta invariant refines the ordinary Zeta polynomial of a finite poset by retaining rank information in weak-chain enumeration [3]. The associated rational generating series has a bivariate numerator, but this numerator need not be coefficientwise nonnegative for an arbitrary ranked poset. For the positive-root poset of type B , Chapoton observed that the coefficients of the powers of t appear to be nonnegative q -analogues of type- B Narayana numbers. With the Lie-rank convention used here, the specialization proved below is $[t^k]\mathbb{H}_{P_r, \text{rk}}(1, t) = \binom{r-1}{k}^2$; equivalently, its Narayana parameter is $r - 1$. The purpose of this paper is to prove an explicit monomial formula that makes this pattern exact, uniformly in the Lie rank. Thus, under the fixed normalization stated below, the paper gives an all-rank proof of the type- B positivity statement highlighted in Chapoton’s discussion and identifies the larger Ferrers-cell mechanism behind it. It is important that the present parameter q records the chain-height weighting in Chapoton’s invariant; it is not the standard q -Narayana statistic studied, for example, in the type- B q -log-convexity literature [6].

2020 *Mathematics Subject Classification.* Primary 06A07; Secondary 05A15, 17B22.

Key words and phrases. root poset, q-Zeta numerator, flag h -vector, EL-labelling, ballot word, lower-semimodular lattice.

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Our contribution is therefore an explicit positive expansion for the q -Zeta numerator itself, together with the EL/ballot mechanism that produces it.

This paper treats the full positive-root posets of standard Lie types B_r and C_r , where r denotes the Lie rank. The rank convention is part of the statement: throughout, the posets are graded by root height minus one and the q -Zeta numerator uses Chapoton's fixed denominator normalization. The fact that the ungraded root posets of types (B) and (C) are isomorphic is known in root-poset combinatorics [8]; here we record an explicit rank-preserving coordinate realisation adapted to this grading in Lemma 3.2.

Let $P_r = \Phi^+(B_r)$ be ordered by nonnegative simple-root difference and graded by

$$\text{rk}(\alpha) = \text{ht}(\alpha) - 1.$$

The q -Zeta numerator $\mathbb{H}_{P_r, \text{rk}}(q, t)$ is defined precisely in Section 2. Put $m = 2r - 2$. Call a word in $\{L, R\}$ a ballot word if every prefix contains at least as many R 's as L 's, and define

$$\mathcal{B}_r = \{w \in \{L, R\}^m : \text{the reversal of } w \text{ is a ballot word}\}.$$

For $w = w_1 \cdots w_m$, let

$$D(w) = \{j : w_j = L \text{ and } (j = 1 \text{ or } w_{j-1} = R)\};$$

thus $D(w)$ records the initial positions of the L -runs. For $r = 1$, these definitions use the empty word. The main statement is the following explicit coefficient formula.

Theorem 1.1. *For every integer $r \geq 1$,*

$$\mathbb{H}_{P_r, \text{rk}}(q, t) = \sum_{w \in \mathcal{B}_r} q^{\sum_{j \in D(w)} (j-1)} t^{|D(w)|}.$$

In particular, every coefficient of the q -Zeta numerator of the standard type- B_r positive-root poset is a nonnegative integer.

The type- A_r companion is the rectangular specialization of the Ferrers theorem (Corollary 5.6), while the identical formula for type C_r follows from the rank-preserving isomorphism in Lemma 3.2; see Corollary 8.2. At $q = 1$, the type- A sequence itself has a rational rank generating function, a second-order recurrence, and strictly interlacing negative zeros (Corollary 5.7); its normalized zero measures converge to the same density as the type- B/C sequence.

Example 1.2. The first three numerators are

$$\mathbb{H}_{P_1, \text{rk}}(q, t) = 1, \quad \mathbb{H}_{P_2, \text{rk}}(q, t) = 1 + t,$$

and

$$\mathbb{H}_{P_3, \text{rk}}(q, t) = 1 + (2 + q + q^2)t + q^2t^2.$$

Thus the formula refines the positivity statement even in small rank: the power of q records the positions at which new left runs begin.

The proof has three structural steps. First, we identify P_r with a trapezoidal cell poset whose order and rank are coordinatewise (Section 3). Second, we show that adjoining a new minimum changes the q -Zeta numerator by the exact substitution $t \mapsto qt$ (Section 6). Third, we equip the augmented cell poset with an EL-labelling by signed endpoint labels. Its maximal chains are precisely the words in $\mathcal{B}_r$, and their descents are the sets $D(w)$ (Sections 7 and 8). We also prove that the augmented poset is a lower-semimodular lattice. For the descent formula we use the explicit EL-labelling directly; no labelling-existence consequence of semimodularity is invoked. Before taking the rank-weighted specialization, the same argument gives a multivariate descent-set polynomial for every Ferrers shape and every interval (Proposition 5.2). The latter polynomial is generated by an explicit finite transfer matrix, so the theorem is also algorithmically effective for arbitrary boundaries (Proposition 5.3). The short criterion in Corollary 6.2 records the general reason that

this construction implies q-Zeta positivity; the Ferrers theorem then supplies a concrete, boundary-parametrized family of such labellings. The formula also yields a coefficientwise shape-monotonicity principle: with top width fixed, relaxing the lower Ferrers boundary enlarges both the global and every common-endpoint local word set. The first t -slice admits a closed q-integer formula, so the initial descent layer reads the Ferrers boundary row by row. At the opposite extreme, the t^{r-1} -slice is a q-generating function of strict row-position sequences and becomes a Gaussian binomial for trapezoidal boundaries. The local interval formula has the same extremal interpretation: its highest descent layer is governed by a flagged strict sequence using only the rows visited by that interval. At fixed top width, every intermediate slice also admits a closed coefficientwise upper envelope from the rectangular Ferrers boundary (Corollary 5.15). For the two-parameter trapezoidal family we additionally give a finite Gaussian-binomial endpoint decomposition for the complete numerator, using the classical one-boundary descent-major-index formula (Theorem 4.2). More strongly, every interval with nonbottom endpoints admits a closed Gaussian-binomial formula depending only on its displacement and boundary slack (Theorem 4.3); at $q = 1$ this becomes an explicit ballot-number difference. At $q = 1$ the whole local t -profile is an explicit ballot-Narayana difference, and it reduces to a binomial product when the boundary is inactive (Corollary 4.6). Principal intervals from the adjoined minimum are handled by a finite sum of the same endpoint contributions, so the trapezoidal lattice has a closed formula on every interval (Theorem 4.5). At $t = 1$ this yields an explicit alternating q-binomial specialization (Corollary 4.7). The first slice gives an exact boundary-deficit formula and shows that this envelope is rigid: equality already forces the boundary to be rectangular (Corollary 5.17). Finally, the same initial-descent analysis determines the characteristic polynomial of every augmented Ferrers lattice: it depends only on the number of rows and the top width, not on the intermediate boundary (Corollary 5.4).

The unrestricted quantifier in Theorem 1.1 is carried entirely by this combinatorial argument. The finite computations described in Section 9.2 are independent consistency checks and are not used to pass from finitely many ranks to all ranks.

Besides positivity, the formula determines the total weight at $(1, 1)$, the sharp t -degree, the leading t -coefficient, and the specialization $[t^k]\mathbb{H}_{P, \text{rk}}(1, t) = \binom{r-1}{k}^2$ (Corollary 8.3). It also gives a sharp degree and unique leading monomial in every fixed t^k -slice (Corollary 8.6). A further flag interpretation gives a three-rank Euler-characteristic refinement (Corollary 8.11). The same $q = 1$ specialization has a stronger spectral consequence: its negative zeros strictly interlace from one Lie rank to the next (Corollary 8.8). The entire rank sequence is governed by the algebraic generating function and three-term recurrence in Corollary 8.9. As the Lie rank tends to infinity, the associated zero measures converge to an explicit density on the negative real axis (Corollary 8.10).

2. PRELIMINARIES AND CONVENTIONS

2.1. Weak chains and the q-Zeta numerator. Let P be a finite graded poset with a unique maximum and rank function $\rho: P \rightarrow \{0, \dots, H\}$. For $m \geq 0$, set

$$A_m(P, \rho; q) = \sum_{x_1 \leq \dots \leq x_m} q^{\rho(x_1) + \dots + \rho(x_m)},$$

where the sum for $m = 0$ consists of the empty chain and equals 1. Chapoton's Lemma 1.1 identifies $A_m(P, \rho; q) = Z_{P, \rho}([m+1]_q)$ for every $m \geq 1$. For the empty-chain term, Chapoton's Lemma 2.2 gives $Z_{P, \rho}([1]_q) = \chi(P) = 1$ whenever P has a unique maximum, so the same indexing also holds for $m = 0$.

Definition 2.1. The weak-chain q-Zeta series is

$$\mathcal{F}_{P, \rho}(q, t) = \sum_{m \geq 0} A_m(P, \rho; q) t^m.$$

In the normalization of [3], its numerator is the polynomial $\mathbb{H}_{P,\rho}(q, t)$ determined by

$$(1) \quad \mathcal{F}_{P,\rho}(q, t) = \frac{\mathbb{H}_{P,\rho}(q, t)}{\prod_{a=0}^H (1 - q^a t)}.$$

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. Thus $\mathbb{H}_{P,\rho} \in \mathbb{N}[q, t]$ means coefficientwise nonnegativity, not merely nonnegativity after specializing q or t .

2.2. Flag numbers and R -labellings. Let Q be a bounded graded poset of ranks $0, \dots, H+1$. For $S \subseteq [H] = \{1, \dots, H\}$, let $\alpha_Q(S)$ be the number of chains whose internal set of ranks is exactly S . Its flag h -number is

$$(2) \quad \beta_Q(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} \alpha_Q(T).$$

Write

$$\sigma(S) = \sum_{s \in S} s.$$

An edge labelling of a bounded graded poset by a totally ordered set is an *EL-labelling* if every interval has a unique maximal chain with strictly increasing labels and this chain is lexicographically first among the maximal chains of that interval. An EL-labelling is, in particular, an *R-labelling* for the usual strict order on labels [2].

The following two standard facts are the only non-elementary poset inputs.

Theorem 2.2 (Chapoton's flag expansion). *For a bounded graded poset Q of ranks $0, \dots, H+1$, equipped with its rank function,*

$$(3) \quad \mathbb{H}_{Q,\text{rk}}(q, t) = \sum_{S \subseteq [H]} \beta_Q(S) q^{\sigma(S)} t^{|S|}.$$

Theorem 2.3 (*R-labellings and descents*). *If a bounded graded poset Q has an R -labelling, then*

$$(4) \quad \beta_Q(S) = \#\{\text{maximal chains of } Q \text{ with descent set } S\}$$

for every set S of internal ranks. In particular, $\beta_Q(S) \geq 0$.

The assertions in Theorems 2.2 and 2.3 are Chapoton's Theorems A.2 and A.1, written in the rank indexing of the present paper: a bounded poset of ranks $0, \dots, H+1$ has internal ranks $[H]$, which is Chapoton's index set $\{1, \dots, H'-1\}$ with $H' = H+1$ [3, Section 5 and Appendix A]. The descent machinery originates in Stanley [12, Theorem 3.1 and Corollary 3.2]; the EL formulation is due to Björner [2].

2.3. Type- B roots. In $\mathbb{R}^r$, with standard basis $e_1, \dots, e_r$, we use the simple roots

$$\alpha_i = e_i - e_{i+1} \quad (1 \leq i < r), \quad \alpha_r = e_r.$$

The positive roots are

$$e_i - e_j \quad (i < j), \quad e_i, \quad e_i + e_j \quad (i < j).$$

For positive roots α, β , the root-poset order is

$$\alpha \leq \beta \iff \beta - \alpha \in \sum_{k=1}^r \mathbb{N} \alpha_k.$$

3. A COMMON CELL MODEL FOR THE TYPE- B AND C ROOT POSETS

Define the trapezoidal set

$$C_r = \{(i, c) : 1 \leq i \leq r, i \leq c \leq 2r - i\}.$$

Give it the order

$$(5) \quad (i, c) \leq (j, d) \iff i \geq j \text{ and } c \leq d.$$

Lemma 3.1. *The map*

$$(6) \quad \begin{aligned} e_i - e_j &\longmapsto (i, j - 1), \\ e_i &\longmapsto (i, r), \\ e_i + e_j &\longmapsto (i, 2r - j + 1) \end{aligned}$$

is a rank-preserving isomorphism from the positive-root poset $P_r = \Phi^+(B_r)$ to C_r . Under this map,

$$\text{rk}(i, c) = c - i.$$

Proof. For fixed i , the three root families in Equation (6) fill, respectively, columns $i, \dots, r - 1$, column r , and columns $r + 1, \dots, 2r - i$. The map is therefore a bijection onto C_r .

We verify the order directly in simple-root coordinates. If $v_{i,c}(k)$ is the coefficient of α_k in the root represented by (i, c) , then for $c \leq r$,

$$v_{i,c}(k) = \begin{cases} 1, & i \leq k \leq c, \\ 0, & \text{otherwise,} \end{cases}$$

whereas for $c > r$, writing $m = 2r - c + 1$,

$$v_{i,c}(k) = \begin{cases} 0, & k < i, \\ 1, & i \leq k < m, \\ 2, & m \leq k \leq r. \end{cases}$$

The cell condition gives $m \geq i + 1$.

Suppose $i \geq j$ and $c \leq d$. If both columns are at most r , the support interval for $v_{i,c}$ is contained in that for $v_{j,d}$. If $c \leq r < d$, the latter vector is at least one on the entire interval from j to r . If $r < c \leq d$, both the support of the positive entries and the terminal interval of entries equal to two for $v_{i,c}$ are contained in the corresponding intervals for $v_{j,d}$. Hence $v_{i,c} \leq v_{j,d}$ coordinatewise.

Conversely, assume $v_{i,c} \leq v_{j,d}$. Comparing the first nonzero coordinate gives $i \geq j$. If $c \leq r$ and $d < r$, then $c > d$ would give $v_{i,c}(c) = 1 > 0 = v_{j,d}(c)$; if $d \geq r$, then $c \leq d$ is automatic. If $c > r$, the coefficient at α_r forces $d > r$. Were $c > d$, the terminal interval of entries equal to two for $v_{i,c}$ would begin strictly earlier than that for $v_{j,d}$, again contradicting coordinatewise domination. Thus $c \leq d$, proving Equation (5).

Finally, the sum of the simple-root coefficients is $c - i + 1$ in both column regions, so root height minus one is $c - i$. $\square$

Lemma 3.2 (Type- C companion cell model). *Let $P_r^C = \Phi^+(C_r)$ be the standard positive-root poset of type C_r , with simple roots $\alpha_i = e_i - e_{i+1}$ for $i < r$ and $\alpha_r = 2e_r$. The map*

$$(7) \quad \begin{aligned} e_i - e_j &\longmapsto (i, j - 1), \\ e_i + e_j &\longmapsto (i, 2r - j), \\ 2e_i &\longmapsto (i, 2r - i) \end{aligned}$$

is a rank-preserving isomorphism from P_r^C to the same trapezoidal cell poset C_r , where the rank is again $c - i$.

Proof. For fixed i , the difference roots fill columns $i, \dots, r-1$, the roots $e_i + e_j$ fill columns $r, \dots, 2r-i-1$, and $2e_i$ fills the last column $2r-i$. Thus the map is bijective. In simple-root coordinates, a cell with $c \leq r$ has coefficients one on $i \leq k \leq c$ and zero elsewhere. For $c > r$, put $m = 2r - c$; its coefficient vector has value zero for $k < i$, value one for $i \leq k < m$, value two for $m \leq k < r$, and value one at $k = r$.

Suppose $i \geq j$ and $c \leq d$. If $d \leq r$, containment of the intervals of ones is immediate. If $c \leq r < d$, the vector for (j, d) is at least one on every coordinate where the vector for (i, c) is nonzero, and it is one at the final coordinate. If $r < c \leq d$, then $2r - d \leq 2r - c$; the support of the ones starts no later and the terminal interval of twos starts no later, so coordinatewise domination again holds. Conversely, coordinatewise domination first gives $i \geq j$ from the first nonzero coordinate. The case $c \leq r$ and $d < r$ then gives $c \leq d$ by looking at coordinate c ; if $d \geq r$ the inequality is automatic. If $c > r$, then $d \leq r$ is impossible because the left vector has a two at coordinate $r-1$ whereas the right vector has at most one. Thus $d > r$, and $c > d$ would make the two-interval of the left vector start strictly earlier, contradicting domination. Hence $c \leq d$.

Finally, the sum of the simple-root coefficients is $c - i + 1$, so height minus one is $c - i$. $\square$

Corollary 3.3. *The poset P_r has ranks $0, \dots, H$, where $H = 2r - 2$, and has a unique maximum, represented by $(1, 2r - 1)$.*

Proof. The minimum possible value of $c - i$ is zero. The largest is $(2r - 1) - 1 = 2r - 2$, attained only at $(1, 2r - 1)$. $\square$

4. A TWO-PARAMETER TRAPEZOIDAL EXTENSION

The preceding argument is not tied to the symmetric boundary of the root poset. Let $s \geq 2r \geq 2$ and define

$$T_{r,s} = \{(i, c) : 1 \leq i \leq r, i \leq c \leq s - i\},$$

ordered by

$$(i, c) \leq (j, d) \iff i \geq j \text{ and } c \leq d,$$

and graded by $\rho(i, c) = c - i$. Adjoin a minimum $\hat{0}$, and give the new cells rank $\hat{\rho}(i, c) = c - i + 1$. Put $m = s - 2$, and let $\mathcal{B}_{r,s}$ consist of the words $w \in \{L, R\}^m$ such that

$$\text{the reversal of } w \text{ is ballot,} \quad \#R(w) - \#L(w) \geq s - 2r.$$

For $w = w_1 \cdots w_m$, retain the run-start set

$$D(w) = \{j : w_j = L \text{ and } (j = 1 \text{ or } w_{j-1} = R)\}.$$

Theorem 4.1 (Trapezoidal-cell q-Zeta formula). *For every $s \geq 2r \geq 2$, the q-Zeta numerator of $T_{r,s}$, with rank $\rho(i, c) = c - i$, is*

$$\mathbb{H}_{T_{r,s}, \rho}(q, t) = \sum_{w \in \mathcal{B}_{r,s}} q^{\sum_{j \in D(w)} (j-1)} t^{|D(w)|}.$$

In particular, it belongs to $\mathbb{N}[q, t]$. For $s = 2r$, one has $\mathcal{B}_{r,2r} = \mathcal{B}_r$ and $T_{r,2r} = C_r$.

Proof of Theorem 4.1. Adjoin a minimum to $T_{r,s}$. For nonbottom cells, the join and meet are

$$(i, c) \vee (j, d) = (\min(i, j), \max(c, d)),$$

and

$$(i, c) \wedge (j, d) = \begin{cases} (\max(i, j), \min(c, d)), & \max(i, j) \leq \min(c, d), \\ \hat{0}, & \max(i, j) > \min(c, d). \end{cases}$$

The inequalities $c \leq s - i$ and $d \leq s - j$ show that the proposed join is a cell, while the same argument for the proposed meet applies when it is nonbottom. The rank is $\widehat{\rho} = c - i + 1$. If the meet is bottom, assume $i \geq j$; then $d < i \leq c$, and

$$\widehat{\rho}(i, c) + \widehat{\rho}(j, d) - \widehat{\rho}(j, c) = d - i + 1 \leq 0.$$

Thus the augmented poset is a lower-semimodular lattice.

Label its covers by

$$\lambda(\widehat{0}, (i, i)) = i, \quad \lambda((i, c), (i - 1, c)) = -(i - 1), \quad \lambda((i, c), (i, c + 1)) = c + 1.$$

On any interval not starting at $\widehat{0}$, all available left labels are negative and occur in increasing order, while all right labels are positive and occur in increasing order. Hence the unique increasing chain performs all left moves before all right moves and is lexicographically first. On an interval from $\widehat{0}$, the increasing chain starts with the atom (j, j) and then uses only right moves. This is an EL-labelling.

A maximal chain starts at (i, i) , uses $i - 1$ left moves and $s - i - 1$ right moves, and ends at $(1, s - 1)$. If L_k, R_k count moves in a prefix, the intermediate cell is $(i - L_k, i + R_k)$, and the upper boundary condition is

$$R_k - L_k \leq s - 2i.$$

At the end equality holds. Consequently the reversed word is ballot, with final balance $b = s - 2i \geq s - 2r$. Conversely, if $w \in \mathcal{B}_{r,s}$ has final balance b , then $b \equiv s - 2 \equiv s \pmod{2}$, so $i = (s - b)/2$ is an integer in $[1, r]$, and the same inequalities keep the path inside $T_{r,s}$. This gives a bijection between maximal chains and $\mathcal{B}_{r,s}$.

The label comparison is identical to the root-poset case: descents occur at rank 1 when the first move is L , and thereafter exactly at positions where R is followed by L . Thus the descent set is $D(w)$. Applying the flag expansion to the EL-labelled augmented lattice and then the exact minimum-adjunction shift from Lemma 6.1 gives the displayed formula. $\square$

4.1. A closed endpoint formula for all t -slices. The ballot-word formula can be converted into a finite Gaussian-binomial expression for the complete numerator, rather than only for its two extreme slices. We use the following standard one-boundary enumerator. If $a, b, d \geq 0$ and $b \leq a + d$, let

$$(8) \quad \Phi_{a,b}^d(x, q) = \sum_{\nu \geq 0} x^\nu q^{\nu^2} \left(\begin{bmatrix} a \\ \nu \end{bmatrix}_q \begin{bmatrix} b \\ \nu \end{bmatrix}_q - q^d \begin{bmatrix} b - d + 1 \\ \nu + 1 \end{bmatrix}_q \begin{bmatrix} a + d - 1 \\ \nu - 1 \end{bmatrix}_q \right).$$

Here a Gaussian binomial is understood to be zero when its lower index is outside the usual range. By the one-boundary descent-major-index formula of Krattenthaler and Mohanty [10, Corollary 2], $\Phi_{a,b}^d(x, q)$ is the generating function

$$\sum_{\pi} x^{\text{des}(\pi)} q^{\text{maj}(\pi)},$$

where π runs over words with a letters L and b letters R whose associated path satisfies $R_k - L_k \leq d$ at every prefix, $\text{des}(\pi) = |\{i : \pi_i = R, \pi_{i+1} = L\}|$, and $\text{maj}(\pi)$ is the sum of the descent positions.

Theorem 4.2 (Gaussian-binomial endpoint decomposition). *Put $m = s - 2$. For $1 \leq a \leq r - 1$, set*

$$b = m - a, \quad d = b - a = m - 2a,$$

and define

$$(9) \quad \mathcal{G}_{a,b}(q, t) = \Phi_{a,b}^d(t, q) + (t - 1)\Phi_{a-1,b}^{d+1}(qt, q).$$

Then

$$(10) \quad \mathbb{H}_{T_{r,s},\rho}(q, t) = 1 + \sum_{a=1}^{r-1} \mathcal{G}_{a,m-a}(q, t).$$

Thus every t -slice of the trapezoidal q -Zeta numerator has an explicit finite Gaussian-binomial expression.

Proof of Theorem 4.2. Fix a word $w \in \mathcal{B}_{r,s}$ with a letters L and $b = m - a$ letters R . Its reversed-ballot condition is equivalent to

$$R_k - L_k \leq b - a = d$$

for every prefix of w , and $a \leq r - 1$. Let

$$S(w) = \{i : w_i = R, w_{i+1} = L\}.$$

The run-start set in the q -Zeta formula satisfies

$$D(w) = (\{1\} \text{ if } w_1 = L, \text{ otherwise } \emptyset) \cup \{i + 1 : i \in S(w)\}.$$

Consequently

$$\sum_{j \in D(w)} (j - 1) = \text{maj}(w), \quad |D(w)| = |S(w)| + \mathbf{1}_{\{w_1=L\}}.$$

The first term in Equation (9) counts all such paths with the ordinary descent weight. For the paths beginning with L , write $w = Lu$. Deleting that initial horizontal step gives a path from the origin to $(a - 1, b)$ under the boundary $R_k - L_k \leq d + 1$. Moreover,

$$\text{des}(w) = \text{des}(u), \quad \text{maj}(w) = \text{maj}(u) + \text{des}(u).$$

Their ordinary descent generating function is therefore $\Phi_{a-1,b}^{d+1}(qt, q)$, and the extra initial run multiplies it by t rather than 1. Replacing the ordinary contribution by this extra factor gives Equation (9). The all- R word contributes the initial 1, and summing over a proves Equation (10).

Finally, Equation (8) is exactly the one-boundary descent-major-index formula cited above, so the expression is finite and explicit. $\square$

Theorem 4.3 (Closed formula for every nonbottom trapezoidal interval). *Let $x = (i, c) \leq y = (j, d)$ be nonbottom cells of $T_{r,s}$, and put*

$$\ell = i - j, \quad u = d - c, \quad \delta = s - i - c.$$

For the induced rank on the interval $[x, y]$, one has

$$(11) \quad \mathbb{H}_{[x,y]}(q, t) = \Phi_{\ell,u}^{\delta}(t, q).$$

In particular,

$$(12) \quad \mathbb{H}_{[x,y]}(1, 1) = \binom{\ell + u}{\ell} - \binom{\ell + u}{\ell + \delta + 1}.$$

Thus every nonbottom interval has a complete q -binomial q -Zeta numerator, depending only on its displacement (ℓ, u) and its boundary slack δ .

Proof of Theorem 4.3. A maximal chain in $[x, y]$ is a word with ℓ letters L and u letters R . After a prefix with counts L_k, R_k , its cell is $(i - L_k, c + R_k)$. The sole nonautomatic cell condition is

$$c + R_k \leq s - (i - L_k), \quad \text{equivalently} \quad R_k - L_k \leq \delta.$$

The local descent set is $\{p : w_p = R, w_{p+1} = L\}$, so its q -Zeta weight is the ordinary descent-major-index weight. Equation (8) therefore gives Equation (11). Setting $x = 1$ and using the one-boundary major-index specialization [10, Corollary 4] yields

$$\Phi_{\ell,u}^\delta(1, q) = \begin{bmatrix} \ell + u \\ \ell \end{bmatrix}_q - q^{\delta+1} \begin{bmatrix} \ell + u \\ \ell + \delta + 1 \end{bmatrix}_q,$$

and $q = 1$ gives Equation (12). $\square$

Definition 4.4 (Endpoint contribution with a prescribed initial atom). For $a, b, d \geq 0$ with $b \leq a + d$, define

$$\Psi_{a,b}^d(q, t) = \begin{cases} 1, & a = 0, \\ \Phi_{a,b}^d(t, q) + (t - 1)\Phi_{a-1,b}^{d+1}(qt, q), & a \geq 1. \end{cases}$$

Theorem 4.5 (Closed formula for every principal trapezoidal interval). Let $y = (j, d) \in T_{r,s}$, put $K = \min\{r, d\}$, and use the induced rank on the principal interval $[\widehat{0}, y]$. Then

$$(13) \quad \mathbb{H}_{[\widehat{0}, y]}(q, t) = \sum_{k=j}^K \Psi_{k-j, d-k}^{s-2k}(q, qt).$$

Consequently, every interval of the augmented trapezoidal lattice has a closed Gaussian-binomial q -Zeta numerator: nonbottom intervals are given by Theorem 4.3, and bottom intervals by Equation (13).

Proof of Theorem 4.5. A maximal chain in $[\widehat{0}, y]$ chooses an initial atom (k, k) with $j \leq k \leq K$, followed by $k - j$ left moves and $d - k$ right moves. For a prefix with L_p, R_p moves, the cell is $(k - L_p, k + R_p)$; the only nonautomatic condition is

$$R_p - L_p \leq s - 2k.$$

The edge from $\widehat{0}$ to (k, k) has label k . It is followed by a descent exactly when the first move is L , while all later descents are the ordinary R -to- L descents. Hence the descent ranks of the chain inside the bounded interval $[\widehat{0}, y]$ form exactly the run-start set $D(w)$ of its move word w , and the flag expansion Theorem 2.2 weights the chain by

$$q^{\sum_{p \in D(w)} p} t^{|D(w)|} = q^{\text{maj}(w)} (qt)^{|D(w)|},$$

using $\sum_{p \in D(w)} (p-1) = \text{maj}(w)$ as in the proof of Theorem 4.2. For fixed k , the deletion argument of Theorem 4.2 evaluates the generating function of $q^{\text{maj}(w)} \tau^{|D(w)|}$ over these words as $\Psi_{k-j, d-k}^{s-2k}(q, \tau)$; when $k = j$ the word is all R and contributes 1. Substituting $\tau = qt$ gives Equation (13). The endpoint inequality $d + j \leq s$ guarantees $d - k \leq (k - j) + (s - 2k)$, so the one-boundary formula applies to every summand. $\square$

Corollary 4.6 (Local $q = 1$ ballot-Narayana profile). With the notation of Theorem 4.3, the complete univariate specialization is

$$(14) \quad \mathbb{H}_{[x, y]}(1, t) = \sum_{\nu \geq 0} t^\nu \left(\binom{\ell}{\nu} \binom{u}{\nu} - \binom{u - \delta + 1}{\nu + 1} \binom{\ell + \delta - 1}{\nu - 1} \right).$$

When the upper boundary is inactive ($\delta \geq u$), this reduces to $\sum_{\nu \geq 0} \binom{\ell}{\nu} \binom{u}{\nu} t^\nu$.

Proof. Set $q = 1$ in Equation (11) and use the definition of $\Phi_{\ell,u}^\delta$. If $\delta \geq u$, the second binomial product vanishes for every ν , giving the stated rectangular specialization. $\square$

Corollary 4.7 (A one-variable q -binomial specialization). *With $m = s - 2$, one has*

$$(15) \quad \mathbb{H}_{T_{r,s},\rho}(q, 1) = 1 + \sum_{a=1}^{r-1} \left(\begin{bmatrix} m \\ a \end{bmatrix}_q - q^{m-2a+1} \begin{bmatrix} m \\ a-1 \end{bmatrix}_q \right).$$

At $q = 1$, this reduces to $\mathbb{H}_{T_{r,s},\rho}(1, 1) = \binom{s-2}{r-1}$.

Proof. Set $t = 1$ in Theorem 4.2; the initial-left-run correction vanishes. The one-boundary major-index specialization [10, Corollary 4] gives

$$\Phi_{a,b}^d(1, q) = \begin{bmatrix} a+b \\ a \end{bmatrix}_q - q^{d+1} \begin{bmatrix} a+b \\ a+d+1 \end{bmatrix}_q.$$

Here $a + b = m$, $d = m - 2a$, and $\begin{bmatrix} m \\ a+d+1 \end{bmatrix}_q = \begin{bmatrix} m \\ m-a-1 \end{bmatrix}_q = \begin{bmatrix} m \\ a-1 \end{bmatrix}_q$. This proves Equation (15). At $q = 1$, the summand is $\binom{m}{a} - \binom{m}{a-1}$, so the sum telescopes. $\square$

Corollary 4.8 (The full $q = 1$ trapezoidal profile). *For $s \geq 2r \geq 2$,*

$$(16) \quad \mathbb{H}_{T_{r,s},\rho}(1, t) = \sum_{k=0}^{r-1} \binom{r-1}{k} \binom{s-r-1}{k} t^k.$$

In particular, this profile is real-rooted with simple negative zeros. More precisely, writing $A = r - 1$ and $B = s - r - 1$, one has

$$\mathbb{H}_{T_{r,s},\rho}(1, t) = (1-t)^A P_A^{(0,B-A)}\left(\frac{1+t}{1-t}\right),$$

where $P_A^{(\alpha,\beta)}$ is the Jacobi polynomial.

Proof. Put $N = s - 2$. For $1 \leq a \leq r - 1$, let $b = N - a$. From Theorem 4.2 and the $q = 1$ specialization of Equation (8), the coefficient of t^k , for $k \geq 1$, in the a -th endpoint contribution is

$$\binom{a-1}{k-1} \binom{N-a+1}{k} - \binom{a}{k} \binom{N-a}{k-1}.$$

The constant term is supplied only by the all- R word. The elementary partial Vandermonde identity

$$\sum_{a=1}^R \left[\binom{a-1}{k-1} \binom{N-a+1}{k} - \binom{a}{k} \binom{N-a}{k-1} \right] = \binom{R}{k} \binom{N-R}{k}$$

(with $R = r - 1$) therefore gives Equation (16). The identity follows by induction on R : the R -th summand is the difference between $\binom{R}{k} \binom{N-R}{k}$ and $\binom{R-1}{k} \binom{N-R+1}{k}$, after two applications of Pascal's identity.

Finally,

$$\sum_{k=0}^A \binom{A}{k} \binom{B}{k} t^k = {}_2F_1(-A, -B; 1; t) = (1-t)^A P_A^{(0,B-A)}\left(\frac{1+t}{1-t}\right).$$

Since $B \geq A$, the Jacobi parameters are nonnegative and its A zeros lie in $(-1, 1)$. The change of variables $t = (x - 1)/(x + 1)$ sends them to simple negative zeros. $\square$

Proposition 4.9 (A rational kernel for the $q = 1$ profiles). *For $A, B \geq 0$, put*

$$J_{A,B}(t) = \sum_{k \geq 0} \binom{A}{k} \binom{B}{k} t^k.$$

Then, as a formal power series,

$$(17) \quad \sum_{A,B \geq 0} J_{A,B}(t) x^A y^B = \frac{1}{(1-x)(1-y) - txy}.$$

Consequently, for $A, B \geq 1$,

$$J_{A,B} = J_{A-1,B} + J_{A,B-1} + (t-1)J_{A-1,B-1},$$

with $J_{A,0} = J_{0,B} = 1$. The trapezoidal profile in Corollary 4.8 is $J_{r-1,s-r-1}(t)$.

Proof. Interchanging the two finite sums with the sums over A and B gives

$$\begin{aligned} \sum_{A,B \geq 0} J_{A,B}(t) x^A y^B &= \sum_{k \geq 0} t^k \frac{x^k}{(1-x)^{k+1}} \frac{y^k}{(1-y)^{k+1}} \\ &= \frac{1}{(1-x)(1-y) - txy}, \end{aligned}$$

using $\sum_{A \geq k} \binom{A}{k} x^A = x^k / (1-x)^{k+1}$. Extracting the coefficient of $x^A y^B$ from $(1-x)(1-y) - txy$ times the series yields the recurrence. $\square$

Proposition 4.10 (Fixed-offset algebraic generating functions). *For the polynomials $J_{A,B}(t)$ in Proposition 4.9, fix $\kappa \in \mathbb{N}$ and put*

$$\Delta_t(z) = \sqrt{(1 + (1-t)z)^2 - 4z}.$$

Then

$$(18) \quad \sum_{n \geq 0} J_{n,n+\kappa}(t) z^n = \frac{1}{\Delta_t(z)} \left(\frac{1 + (1-t)z - \Delta_t(z)}{2z} \right)^\kappa$$

as an identity of formal power series in z with coefficients in $\mathbb{Q}[t]$. In particular, for every real $t \geq 0$ and every fixed $\kappa \in \mathbb{N}$,

$$(19) \quad \lim_{n \rightarrow \infty} J_{n,n+\kappa}(t)^{1/n} = (1 + \sqrt{t})^2.$$

Proof. Set $a = 1 + (1-t)z$. Extracting the coefficient of $x^{-\kappa}$ from the rational kernel after the substitution $y = z/x$ gives

$$z^\kappa \sum_{n \geq 0} J_{n,n+\kappa}(t) z^n = [x^{-\kappa}] \frac{1}{a - x - z/x}.$$

Expanding formally in $(x + z/x)/a$ yields

$$[x^{-\kappa}] \frac{1}{a - x - z/x} = z^\kappa \sum_{j \geq 0} \binom{2j + \kappa}{j} \frac{z^j}{a^{2j + \kappa + 1}}.$$

The standard Catalan-series identity

$$\sum_{j \geq 0} \binom{2j + \kappa}{j} u^j = \frac{C(u)^\kappa}{\sqrt{1 - 4u}}, \quad C(u) = \frac{1 - \sqrt{1 - 4u}}{2u},$$

which follows from the Catalan equation $C(u) = 1 + uC(u)^2$ by coefficient extraction; with $u = z/a^2$, it gives (18).

For the growth statement, coefficientwise monotonicity and

$$1 \leq \frac{\binom{n+\kappa}{k}}{\binom{n}{k}} \leq \binom{n+\kappa}{\kappa} \quad (0 \leq k \leq n)$$

give

$$J_{n,n}(t) \leq J_{n,n+\kappa}(t) \leq \binom{n+\kappa}{\kappa} J_{n,n}(t).$$

Finally, if $c_k = \binom{n}{k} t^{k/2}$, then Cauchy–Schwarz gives

$$\frac{(1 + \sqrt{t})^{2n}}{n + 1} \leq J_{n,n}(t) \leq (1 + \sqrt{t})^{2n}.$$

Taking n th roots proves (19). $\square$

Corollary 4.11 (Fixed-offset Sturm interlacing). *Fix $\kappa \in \mathbb{N}$, and for $n \geq 1$ write $J_{n,n+\kappa}(t) = \prod_{j=1}^n (t - \zeta_{n,\kappa,j})$ up to its positive leading factor, with $\zeta_{n,\kappa,1} < \dots < \zeta_{n,\kappa,n} < 0$. Then the zeros of $J_{n,n+\kappa}$ and $J_{n+1,n+1+\kappa}$ strictly interlace on $(-\infty, 0)$:*

$$\zeta_{n+1,\kappa,1} < \zeta_{n,\kappa,1} < \zeta_{n+1,\kappa,2} < \dots < \zeta_{n,\kappa,n} < \zeta_{n+1,\kappa,n+1}.$$

Proof. The Jacobi representation in Corollary 4.8 gives

$$J_{n,n+\kappa}(t) = (1 - t)^n P_n^{(0,\kappa)}\left(\frac{1+t}{1-t}\right).$$

For fixed nonnegative parameters, consecutive Jacobi polynomials have strictly interlacing zeros in $(-1, 1)$ [14, Chapter 6]. The change of variables $t = (x - 1)/(x + 1)$ is strictly increasing from $(-1, 1)$ onto $(-\infty, 0)$, so it preserves the interlacing order. $\square$

Corollary 4.12 (Fixed-offset asymptotic zero law). *Fix $\kappa \in \mathbb{N}$, and for $r \geq 2$ set $s = 2r + \kappa$. If $\tau_{r,\kappa,1} < \dots < \tau_{r,\kappa,r-1} < 0$ are the zeros of $\mathbb{H}_{T_{r,2r+\kappa,\rho}}(1, t)$, then for every continuous compactly supported f on $(-\infty, 0)$,*

$$\lim_{r \rightarrow \infty} \frac{1}{r-1} \sum_{j=1}^{r-1} f(\tau_{r,\kappa,j}) = \frac{1}{\pi} \int_{-\infty}^0 \frac{f(t)}{\sqrt{-t}(1-t)} dt.$$

The limit is independent of κ . Equivalently, if $j_r/(r-1) \rightarrow u \in (0, 1)$, then

$$\tau_{r,\kappa,j_r} \rightarrow -\cot^2\left(\frac{\pi u}{2}\right).$$

Proof. For $n = r - 1$, the preceding corollary gives

$$\mathbb{H}_{T_{r,2r+\kappa,\rho}}(1, t) = (1 - t)^n P_n^{(0,\kappa)}\left(\frac{1+t}{1-t}\right).$$

The normalized zero measures of $P_n^{(0,\kappa)}$ converge to the arcsine measure on $(-1, 1)$ for every fixed κ ; this is the standard fixed-parameter Jacobi zero law [14]. Pushing it forward under $t = (x-1)/(x+1)$ gives the displayed density. Its distribution function is $F(t) = \frac{2}{\pi} \arctan(1/\sqrt{-t})$, so the quantile statement follows. $\square$

Corollary 4.13 (Sharp slices for the trapezoidal family). *For $0 \leq k \leq r - 1$,*

$$[t^k] \mathbb{H}_{T_{r,s,\rho}}(q, t) \in q^{k(k-1)} \mathbb{N}[q], \quad \deg_q[t^k] \mathbb{H}_{T_{r,s,\rho}}(q, t) = k(s - k - 3).$$

The coefficient of $q^{k(s-k-3)}$ is 1.

Proof. For a word w with $D(w) = \{j_1 < \dots < j_k\}$, consecutive run starts are separated by a right letter, so $j_a \geq 2a - 1$. Hence

$$\sum_{a=1}^k (j_a - 1) \geq k(k - 1).$$

Let v be the reversal of w , and let $p_1 < \dots < p_k$ be the reversed positions corresponding to the elements of $D(w)$. Each p_a is an L -position of the ballot word v . Among its first p_a letters there

are at least a such positions, and the ballot condition forces at least a R 's there; thus $p_a \geq 2a$. Writing $m = s - 2$,

$$\sum_{a=1}^k (j_a - 1) = \sum_{a=1}^k (m - p_a) \leq \sum_{a=1}^k (m - 2a) = k(s - k - 3).$$

Equality forces $p_a = 2a$ for every a . The ballot condition then forces v to begin with $(RL)^k$; since there are exactly k marked positions, all remaining letters are R . Therefore $v = (RL)^k R^{s-2-2k}$, which is admissible precisely when $k \leq r - 1$. This unique word attains the upper bound. $\square$

Corollary 4.14 (Specialization of the trapezoidal family). *For $s \geq 2r \geq 2$,*

$$\mathbb{H}_{T_{r,s,\rho}}(1, 1) = \binom{s-2}{r-1}, \quad \deg_t \mathbb{H}_{T_{r,s,\rho}} = r - 1,$$

and

$$[t^{r-1}] \mathbb{H}_{T_{r,s,\rho}}(1, t) = \binom{s-r-1}{r-1}.$$

Proof. Write $m = s - 2$. A ballot word with exactly ℓ letters L is counted by the reflection principle as $\binom{m}{\ell} - \binom{m}{\ell-1}$. The defining final-balance condition for $\mathcal{B}_{r,s}$ is equivalent to $\ell \leq r - 1$. Summing over ℓ telescopes to $\binom{m}{r-1}$, proving the first identity. A word contributes to t^{r-1} only when $\ell = r - 1$ and every L lies in its own run. Write such a word as $R^{a_0} L R^{a_1} L \cdots L R^{a_{r-1}}$ with cumulative sums $A_u = a_0 + \cdots + a_u$: isolation of the runs means $a_1, \dots, a_{r-2} \geq 1$, the reversed-ballot condition is exactly $A_u \leq (s - 2r) + u$ for $0 \leq u \leq r - 2$, and the final run length a_{r-1} is then determined and positive. The substitution $C_u = A_u - u$ identifies these data with the weakly increasing sequences $0 \leq C_0 \leq \cdots \leq C_{r-2} \leq s - 2r$, of which there are $\binom{s-r-1}{r-1}$. This proves the third identity; since $s \geq 2r$ makes this count positive and no word has more than $r - 1$ descents, the t -degree is exactly $r - 1$. $\square$

5. FERRERS-CELL UNIVERSALITY

The trapezoidal boundary is only one instance of a monotone Ferrers boundary. Let $r \geq 1$ and let

$$b_1 \geq b_2 \geq \cdots \geq b_r, \quad b_i \geq i,$$

be integers. Define

$$F_{\mathbf{b}} = \{(i, c) : 1 \leq i \leq r, i \leq c \leq b_i\},$$

with the same order

$$(i, c) \leq (j, d) \iff i \geq j \text{ and } c \leq d,$$

and rank $\rho(i, c) = c - i$. Put $M = b_1 - 1$. For a word $w \in \{L, R\}^M$, write $\ell(w)$ for its number of L 's and set $\iota(w) = \ell(w) + 1$. Let $\mathcal{W}_{\mathbf{b}}$ consist of the words with $\ell(w) \leq r - 1$ such that, for every prefix with L_k letters L and R_k letters R ,

$$\iota(w) + R_k \leq b_{\iota(w) - L_k}.$$

Theorem 5.1 (Ferrers-cell q-Zeta formula). *For every boundary sequence $\mathbf{b}$ as above,*

$$\mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = \sum_{w \in \mathcal{W}_{\mathbf{b}}} q^{\sum_{j \in D(w)} (j-1)} t^{|D(w)|}.$$

In particular, $\mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) \in \mathbb{N}[q, t]$. The trapezoidal family $T_{r,s}$ is obtained by $b_i = s - i$.

Proof of Theorem 5.1. Adjoin a minimum to $F_{\mathbf{b}}$. The same coordinate formulas give

$$(i, c) \vee (j, d) = (\min(i, j), \max(c, d))$$

and

$$(i, c) \wedge (j, d) = \begin{cases} (\max(i, j), \min(c, d)), & \max(i, j) \leq \min(c, d), \\ \widehat{0}, & \text{otherwise.} \end{cases}$$

The monotonicity of $\mathbf{b}$ makes the displayed join and every nonbottom meet a cell. Label the covers by

$$\lambda(\widehat{0}, (i, i)) = i, \quad \lambda((i, c), (i - 1, c)) = -(i - 1), \quad \lambda((i, c), (i, c + 1)) = c + 1.$$

The covers are exactly the displayed left and right moves, and each increases $\rho(i, c) = c - i$ by one; thus ρ is a rank function on $F_{\mathbf{b}}$. On an interval not starting at the minimum, all left labels are negative and increase as the first coordinate decreases, while all right labels are positive and increase as the second coordinate increases. Hence the unique increasing chain takes all available left moves before all right moves and is lexicographically first. On an interval from the minimum to (j, d) , the increasing chain starts with the least-labelled available atom (j, j) , followed only by right moves. Thus λ is an EL-labelling.

A maximal chain starts at (i, i) , makes $i - 1$ left moves and $b_1 - i$ right moves, and ends at $(1, b_1)$. Its move word has length M , and the cell after a prefix is $(i - L_k, i + R_k)$. Therefore the path stays in $F_{\mathbf{b}}$ exactly when the defining inequalities for $\mathcal{W}_{\mathbf{b}}$ hold, with $i = \iota(w)$. This gives a bijection between maximal chains and $\mathcal{W}_{\mathbf{b}}$. The comparison of the initial atom label, negative left labels, and positive right labels is unchanged from the trapezoidal case, so the descent set is $D(w)$. The EL flag expansion and the minimum-adjunction shift now give the displayed formula. $\square$

Proposition 5.2 (Multivariate descent-set refinement). *Let $M = b_1 - 1$ and let $\mathbf{u} = (u_1, \dots, u_M)$ be commuting indeterminates. The multivariate descent enumerator*

$$\mathcal{D}_{\mathbf{b}}(\mathbf{u}) = \sum_{w \in \mathcal{W}_{\mathbf{b}}} \prod_{j \in D(w)} u_j$$

is coefficientwise positive, and the q -Zeta numerator is its principal specialization

$$(20) \quad \mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = \mathcal{D}_{\mathbf{b}}(t, tq, tq^2, \dots, tq^{M-1}).$$

More generally, for a nonbottom interval $I = [x, y]$ of length $h \geq 1$, the local descent enumerator

$$\mathcal{D}_I(\mathbf{u}) = \sum_{w \in \mathcal{W}_{\mathbf{b}}(x, y)} \prod_{p \in D_{x, y}(w)} u_p$$

specializes to

$$\mathbb{H}_{I, \rho_{x, y}}(q, t) = \mathcal{D}_I(tq, tq^2, \dots, tq^{h-1}).$$

Thus the word model retains the complete flag descent-set data, not only its rank-weighted q -Zeta specialization.

Proof of Proposition 5.2. The first assertion is immediate from the definition: each admissible word contributes one monomial with coefficient one. In the global formula of Theorem 5.1, substituting $u_j = tq^{j-1}$ gives

$$\prod_{j \in D(w)} u_j = t^{|D(w)|} q^{\sum_{j \in D(w)} (j-1)},$$

which proves Equation (20). For an interval, the same substitution with $u_p = tq^p$ and the local word formula of Proposition 5.10 gives the final identity. $\square$

Proposition 5.3 (Transfer-matrix recursion). *Let $M = b_1 - 1$, and let $\sigma \in \{\emptyset, L, R\}$ record the previous move. Define polynomials $G_j(i, c; \sigma)$ recursively by $G_0(i, i; \emptyset) = 1$ for $1 \leq i \leq r$, with all other initial values zero, and, for $1 \leq j \leq M$, by the transitions*

$$\begin{aligned} G_j(i-1, c; L) &+= G_{j-1}(i, c; \sigma) \cdot u_j^{\mathbf{1}_{\{\sigma \in \{\emptyset, R\}\}}}, & (i-1, c) \in F_{\mathbf{b}}, \\ G_j(i, c+1; R) &+= G_{j-1}(i, c; \sigma), & (i, c+1) \in F_{\mathbf{b}}, \end{aligned}$$

for each σ . Then

$$\mathcal{D}_{\mathbf{b}}(\mathbf{u}) = \sum_{\sigma \in \{\emptyset, L, R\}} G_M(1, b_1; \sigma).$$

In particular, the complete multivariate enumerator, and hence the q -Zeta numerator by Equation (20), is computable with $O(M|F_{\mathbf{b}}|)$ polynomial-ring transitions.

Proof of Proposition 5.3. After j transitions, $G_j(i, c; \sigma)$ is the sum of the monomials attached to all legal length- j paths ending at (i, c) whose last move is σ . A left transition creates a descent exactly when it is the first move or follows a right move, which is precisely the exponent in the first line; right transitions never create a descent. Induction on j proves the state interpretation. Every maximal path has length M and ends at $(1, b_1)$, so summing the three terminal states gives the claimed enumerator. There are $O(M|F_{\mathbf{b}}|)$ states and a bounded number of transitions per state. $\square$

Corollary 5.4 (Characteristic polynomial of the augmented Ferrers lattice). *Let $\widehat{F}_{\mathbf{b}} = F_{\mathbf{b}} \sqcup \{\widehat{0}\}$, with the shifted rank from Lemma 6.1, and let*

$$X_{\widehat{F}_{\mathbf{b}}}(y) = \sum_{x \in \widehat{F}_{\mathbf{b}}} \mu(\widehat{0}, x) y^{b_1 - \widehat{\rho}(x)}$$

be its characteristic polynomial. Then

$$(21) \quad X_{\widehat{F}_{\mathbf{b}}}(y) = y^{b_1} - ry^{b_1-1} + (r-1)y^{b_1-2},$$

where the last term is understood to be absent when $r = 1$. Thus the characteristic polynomial is independent of all lower-boundary entries $b_2, \dots, b_r$. In particular, for the balanced boundary $b_i = 2r - i$ and $r \geq 2$,

$$X_{\widehat{C}_r}(y) = y^{2r-3}(y-1)(y-(r-1)),$$

where $\widehat{C}_r$ denotes the minimum-augmented type- B_r/C_r root-poset lattice.

Proof. The augmented lattice has maximal rank b_1 . By the Theorem 2.3, its flag h -number $\beta(S)$ counts maximal chains with descent set S . The ballot-word description in Theorem 5.1 shows that $\beta(\emptyset) = 1$ and $\beta(\{1\}) = r - 1$: the latter are exactly the chains that start with a left move and then make all remaining left moves before the right moves. No descent set can contain both 1 and 2, and hence $\beta(\{1, \dots, j\}) = 0$ for every $j \geq 2$. Applying Chapoton's $q = 0$ characteristic-polynomial identity [3, Lemma 6.1 and Theorem 6.2] together with its flag-vector form gives

$$y^{b_1} X_{\widehat{F}_{\mathbf{b}}}(1/y) = 1 - ry + (r-1)y^2.$$

Replacing y by $1/y$ and multiplying by y^{b_1} yields Equation (21). $\square$

Corollary 5.5 (Characteristic polynomials of principal ideals). *For a cell $y = (j, d) \in F_{\mathbf{b}}$, let*

$$h_y = d - j + 1, \quad a_y = \min\{r, d\} - j + 1.$$

Then the principal ideal $[\widehat{0}, y] \subseteq \widehat{F}_{\mathbf{b}}$ has characteristic polynomial

$$X_{[\widehat{0}, y]}(z) = z^{h_y} - a_y z^{h_y-1} + (a_y - 1) z^{h_y-2},$$

where the final term is absent when $a_y = 1$. In particular, this polynomial is independent of the intermediate boundary values met below y .

Proof. The interval $[\widehat{0}, y]$ inherits the signed EL-labelling. Its unique increasing maximal chain starts at the least-labelled atom (j, j) and then uses right moves, so its empty descent set occurs once. The atoms of the interval are exactly (i, i) with $j \leq i \leq \min\{r, d\}$, hence there are a_y of them. A maximal chain has descent set $\{1\}$ precisely when it starts at one of the other $a_y - 1$ atoms and makes all its left moves before its right moves. No descent set contains both 1 and 2, so all initial descent sets $\{1, \dots, k\}$ with $k \geq 2$ have zero flag h -number. Chapoton's $q = 0$ characteristic-polynomial identity [3, Lemma 6.1 and Theorem 6.2] now gives

$$z^{h_y} X_{[\widehat{0}, y]}(1/z) = 1 - a_y z + (a_y - 1)z^2,$$

which is equivalent to the displayed formula. $\square$

Corollary 5.6 (Type-A companion formula). *Let $P_r^A = \Phi^+(A_r)$ be the standard positive-root poset of type A_r , graded by root height minus one. Then*

$$\mathbb{H}_{P_r^A, \rho}(q, t) = \sum_{w \in \{L, R\}^{r-1}} q^{\sum_{j \in D(w)} (j-1)} t^{|D(w)|}.$$

In particular, its q -Zeta numerator is coefficientwise nonnegative for every $r \geq 1$, and for $1 \leq k \leq \lfloor r/2 \rfloor$,

$$[t^k] \mathbb{H}_{P_r^A, \rho}(q, t) = q^{k(k-1)} \sum_{h=k}^{r-k} \begin{bmatrix} h-1 \\ k-1 \end{bmatrix}_q \begin{bmatrix} r-h \\ k \end{bmatrix}_q.$$

The coefficients vanish for $k > \lfloor r/2 \rfloor$, while the constant coefficient is 1. At $q = 1$, the convolution reduces to

$$\mathbb{H}_{P_r^A, \rho}(1, t) = \sum_{k=0}^{\lfloor r/2 \rfloor} \binom{r}{2k} t^k = \frac{(1 + \sqrt{t})^r + (1 - \sqrt{t})^r}{2}.$$

For $r \geq 2$, this polynomial has exactly $\lfloor r/2 \rfloor$ distinct negative real zeros.

Proof. Write the positive roots of A_r as $\alpha_{i,j} = e_i - e_{j+1}$ with $1 \leq i \leq j \leq r$. The map $\alpha_{i,j} \mapsto (i, j)$ identifies the root order with

$$(i, c) \leq (j, d) \iff i \geq j, c \leq d$$

on $F_{\bar{\mathbf{b}}}$, where $\bar{\mathbf{b}} = (r, \dots, r)$, and it preserves the rank $\text{ht}(\alpha_{i,j}) - 1 = j - i$. The Ferrers theorem therefore gives the word formula. For this constant boundary every word of length $r-1$ is admissible. A binary word of length $r-1$ with k L -runs is specified by the $2k$ alternating endpoints of those runs among the r gaps, so there are $\binom{r}{2k}$ such words. Summing over k and applying the binomial theorem yields the final identity. The displayed coefficient convolution is precisely the rectangular case of Corollary 5.15; at $q = 1$, Vandermonde's identity gives $\binom{r}{2k}$. To locate the zeros, put $x = \sqrt{t}$. Apart from the exceptional value $\zeta = -1$, the equation $(1+x)^r + (1-x)^r = 0$ is equivalent to

$$\left(\frac{1+x}{1-x} \right)^r = -1.$$

Thus $x = (\zeta - 1)/(\zeta + 1)$, where ζ runs over the roots of $\zeta^r = -1$ with $\zeta \neq -1$. Since $|\zeta| = 1$, each such x is purely imaginary; conjugate values of ζ give the same negative value of $t = x^2$, and no other identifications occur. The resulting number of distinct roots is $\lfloor r/2 \rfloor$, as claimed. $\square$

Corollary 5.7 (Type-A rank spectrum). *Write*

$$A_r(t) = \mathbb{H}_{P_r^A, \rho}(1, t) = \sum_{k=0}^{\lfloor r/2 \rfloor} \binom{r}{2k} t^k \quad (r \geq 1),$$

and set $A_0(t) = 1$. Then

$$\sum_{r \geq 0} A_r(t) z^r = \frac{1-z}{1-2z+(1-t)z^2},$$

so that $A_1(t) = 1$, $A_2(t) = 1 + t$, and

$$A_r(t) = 2A_{r-1}(t) - (1-t)A_{r-2}(t) \quad (r \geq 2).$$

Moreover, the negative zeros of A_r and A_{r+1} strictly interlace on $(-\infty, 0)$, in the usual sense for polynomials whose degrees differ by at most one. If $\xi_{r,1}, \dots, \xi_{r,d_r}$ are the zeros, where $d_r = \lfloor r/2 \rfloor$, then their normalized zero-counting measures have the same limit as in the type-B/C case: for every $f \in C_c((-\infty, 0))$,

$$\lim_{r \rightarrow \infty} \frac{1}{d_r} \sum_{j=1}^{d_r} f(\xi_{r,j}) = \frac{1}{\pi} \int_{-\infty}^0 \frac{f(t)}{\sqrt{-t}(1-t)} dt.$$

When the zeros are ordered increasingly and $j_r/d_r \rightarrow u \in (0, 1)$,

$$\xi_{r,j_r} \rightarrow -\cot^2\left(\frac{\pi u}{2}\right).$$

Proof. The closed form in Corollary 5.6 gives

$$A_r(t) = \frac{(1 + \sqrt{t})^r + (1 - \sqrt{t})^r}{2}.$$

Summing the two geometric series proves the displayed generating function and its recurrence. For the zero statement, put $d_r = \lfloor r/2 \rfloor$. The finite zeros are

$$\xi_{r,j} = -\tan^2\left(\frac{(2j-1)\pi}{2r}\right), \quad 1 \leq j \leq d_r.$$

Indeed, the equation $A_r(t) = 0$, with $x = \sqrt{t}$, is equivalent to

$$\left(\frac{1+x}{1-x}\right)^r = -1,$$

and the nonreal unit roots of the right-hand side give the displayed purely imaginary x 's and hence the negative values $\xi_{r,j}$. The angles $(2j-1)\pi/r$ for rank r and rank $r+1$ alternate. If $r = 2m$, then

$$\frac{(2j-1)\pi}{2m+1} < \frac{(2j-1)\pi}{2m} < \frac{(2j+1)\pi}{2m+1} \quad (1 \leq j < m),$$

and the final rank- $(2m+1)$ angle precedes the final rank- $2m$ angle. If $r = 2m+1$, the same inequalities with denominators $2m+1$ and $2m+2$ hold for $1 \leq j \leq m$, and there is one additional rank- $(2m+2)$ angle at the endpoint. Since $-\tan^2(\theta/2)$ is strictly decreasing on $(0, \pi)$, the negative zeros alternate as claimed. Finally, the angles $\theta_{r,j} = (2j-1)\pi/(2r)$ form a midpoint Riemann grid on $(0, \pi/2)$, and $d_r/r \rightarrow 1/2$. Hence

$$\frac{1}{d_r} \sum_{j=1}^{d_r} f(\xi_{r,j}) \rightarrow \frac{2}{\pi} \int_0^{\pi/2} f(-\tan^2 \theta) d\theta.$$

The substitution $t = -\tan^2 \theta$ gives the displayed density. Its distribution function is $F(t) = \frac{2}{\pi} \arctan(1/\sqrt{-t})$, so quantile convergence yields the final formula. $\square$

Remark 5.8 (The mechanism is genuinely non-distributive). For $r \geq 3$, the augmented Ferrers-cell lattice is not distributive. Indeed, with $x = (2, 2)$, $y = (1, 1)$, and $z = (r, r)$, one has

$$x \wedge (y \vee z) = x, \quad (x \wedge y) \vee (x \wedge z) = \widehat{0}.$$

Thus the positivity theorem is not a disguised application of the standard distributive-lattice criterion; it uses the explicit signed EL-labelling.

Corollary 5.9 (Interval-stable positivity). *Let $I = [x, y]$ be any interval of the augmented Ferrers-cell lattice $\widehat{F}_{\mathbf{b}} = F_{\mathbf{b}} \sqcup \{\widehat{0}\}$, and give I the induced rank function $\widehat{\rho}_I(z) = \widehat{\rho}(z) - \widehat{\rho}(x)$. Then*

$$\mathbb{H}_{I, \widehat{\rho}_I}(q, t) \in \mathbb{N}[q, t].$$

Proof. Restrict the signed labelling in the proof of Theorem 5.1 to the interval I . The defining increasing chain and lexicographic-first property are intervalwise, so the restriction is an EL-labelling, hence an R -labelling, of the bounded graded poset I . Applying the flag expansion Theorem 2.2 to I and evaluating its flag h -numbers by Theorem 2.3 writes $\mathbb{H}_{I, \widehat{\rho}_I}$ as a sum of monomials $q^{\sigma(S)} t^{|S|}$ with nonnegative multiplicities. $\square$

Proposition 5.10 (Local ballot-word formula). *Let $x = (i, c) \leq y = (j, d)$ be nonbottom cells of $F_{\mathbf{b}}$, and put $h = (i - j) + (d - c)$. Let $\mathcal{W}_{\mathbf{b}}(x, y)$ be the words with exactly $i - j$ letters L and $d - c$ letters R such that, for every prefix with counts L_k, R_k ,*

$$c + R_k \leq b_{i-L_k}.$$

For such a word set

$$D_{x,y}(w) = \{p \in \{1, \dots, h-1\} : w_p = R, w_{p+1} = L\}.$$

Then, with the induced rank $\rho_{x,y}(z) = \rho(z) - \rho(x)$,

$$\mathbb{H}_{[x,y], \rho_{x,y}}(q, t) = \sum_{w \in \mathcal{W}_{\mathbf{b}}(x,y)} q^{\sum_{p \in D_{x,y}(w)} p} t^{|D_{x,y}(w)|}.$$

Proof of Proposition 5.10. Maximal chains in $[x, y]$ are precisely the legal shuffles of the $i - j$ left moves and $d - c$ right moves. After a prefix they are at $(i - L_k, c + R_k)$, so legality is exactly the displayed boundary condition. The edge labels along consecutive left moves and consecutive right moves are strictly increasing; a left move followed by a right move is increasing, while a right move followed by a left move is a descent. Thus the descent set is $D_{x,y}(w)$. Apply the bounded flag expansion and the R -labelling descent theorem to the interval. $\square$

Corollary 5.11 (Top slice of a nonbottom interval). *Let $x = (i, c) \leq y = (j, d)$ be nonbottom cells and put $\ell = i - j$ and $R = d - c$. For $0 \leq u \leq \ell - 1$, define*

$$B_u = \min\{R, b_{i-u} - c\}.$$

Then

$$[t^\ell] \mathbb{H}_{[x,y], \rho_{x,y}}(q, t) = q^{\binom{\ell}{2}} \sum_{\substack{1 \leq A_0 < \dots < A_{\ell-1} \\ A_u \leq B_u}} q^{A_0 + \dots + A_{\ell-1}}.$$

For $\ell = 0$, the sum consists of the single empty sequence and equals 1. Thus the highest descent layer of every nonbottom interval is a flagged strict-sequence enumerator determined only by the boundary rows met by the interval and its horizontal displacement R .

Proof. In the local descent set $D_{x,y}(w)$, a left move is counted exactly when it immediately follows a right move. A word contributing to t^ℓ therefore has each of its ℓ left moves in a separate run and begins with a right move. It thus has the form

$$R^{a_0} L R^{a_1} L \dots L R^{a_{\ell-1}} L R^{a_\ell},$$

where $a_0 \geq 1$, $a_u \geq 1$ for $1 \leq u < \ell$, $a_\ell \geq 0$, and $a_0 + \dots + a_\ell = R$. Set $A_u = a_0 + \dots + a_u$ for $0 \leq u \leq \ell - 1$; these are strictly increasing with $A_0 \geq 1$. The internal right-run conditions are exactly $A_u \leq b_{i-u} - c$, while the final run condition is $A_u \leq R$. Hence the admissible run data are precisely the displayed strict sequences. The $(u+1)$ -st left letter is at position $A_u + u + 1$, so the descent preceding it is at position $A_u + u$, and the local formula of Proposition 5.10 weights each word by q to the sum of these positions times t^ℓ . Summing the descent positions gives $\binom{\ell}{2} + \sum_u A_u$, proving the formula. $\square$

Corollary 5.12 (Gaussian-binomial determinant for the local top slice). *With the notation of Corollary 5.11, assume $\ell \geq 1$ and use the convention $\begin{bmatrix} N \\ k \end{bmatrix}_q = 0$ for $k < 0$ or $N < k$. Then*

$$[t^\ell] \mathbb{H}_{[x,y], \rho_{x,y}}(q, t) = q^{\binom{\ell+1}{2}} \det_{1 \leq a, b \leq \ell} \begin{bmatrix} B_{a-1} - a + b \\ 1 - a + b \end{bmatrix}_q.$$

At $q = 1$ this specializes to the flagged-column determinant in ordinary binomial coefficients. For a trapezoidal interval, writing $\delta = s - i - c$ gives $B_u = \min\{R, \delta + u\}$; when $R \geq \delta + \ell - 1$, the slice reduces to $q^{\ell^2} \begin{bmatrix} \delta + \ell - 1 \\ \ell \end{bmatrix}_q$, and hence to $\binom{\delta + \ell - 1}{\ell}$ at $q = 1$.

Proof. The preceding corollary counts strictly increasing sequences $1 \leq A_0 < \dots < A_{\ell-1}$ with $A_u \leq B_u$, weighted by $q^{A_0 + \dots + A_{\ell-1}}$. Subtracting 1 from each term costs the factor q^ℓ and leaves the sequences $0 \leq A'_0 < \dots < A'_{\ell-1}$ with flags $A'_u \leq B_u - 1$. A standard flagged-Schur principal specialization [16], equivalently the Lindström–Gessel–Viennot path expansion, counts the latter as

$$\det_{1 \leq a, b \leq \ell} \begin{bmatrix} (B_{a-1} - 1) + 1 - a + b \\ 1 - a + b \end{bmatrix}_q = \det_{1 \leq a, b \leq \ell} \begin{bmatrix} B_{a-1} - a + b \\ 1 - a + b \end{bmatrix}_q.$$

Combining q^ℓ with the descent-position offset $q^{\binom{\ell}{2}}$ from Corollary 5.11 gives the outer factor $q^{\binom{\ell+1}{2}}$. For a trapezoidal boundary, $B_u = \min\{R, \delta + u\}$ as above; if $R \geq \delta + \ell - 1$, the shifted sequences are exactly those with $A'_u \leq \delta + u - 1$, and the substitution $C_u = A'_u - u$ turns them into partitions inside an $\ell \times (\delta - 1)$ box, with generating function $q^{\binom{\ell}{2}} \begin{bmatrix} \delta + \ell - 1 \\ \ell \end{bmatrix}_q$. The total q -exponent is then $\binom{\ell+1}{2} + \binom{\ell}{2} = \ell^2$. $\square$

Corollary 5.13 (A boundary-independent $q = 0$ edge). *For every boundary $\mathbf{b}$ as above,*

$$\mathbb{H}_{F_{\mathbf{b}}, \rho}(0, t) = 1 + (r - 1)t.$$

For every interval $[x, y]$ with nonbottom lower endpoint, $\mathbb{H}_{[x,y], \rho_{x,y}}(0, t) = 1$.

Proof. In the global word formula, a monomial survives at $q = 0$ only when its descent set is empty or is $\{1\}$. The unique word with empty descent set is R^{b_1-1} . For each $1 \leq a \leq r - 1$, the word $L^a R^{b_1-1-a}$ is admissible and has descent set $\{1\}$, and these are all possibilities. This gives the first identity. In the local interval formula the exponent is the sum of positive internal descent positions, so only the empty descent set remains. Its unique word is the all-left-then-right shuffle $L^{i-j} R^{d-c}$, which is legal because the boundary is nonincreasing. Hence the local numerator is 1. $\square$

Corollary 5.14 (Shape monotonicity). *Fix r and b_1 . Let $\mathbf{b} = (b_1, \dots, b_r)$ and $\mathbf{b}' = (b'_1, \dots, b'_r)$ be two nonincreasing boundary sequences with $b'_1 = b_1$ and $b_i \leq b'_i$ for every i . Then*

$$\mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) \preceq \mathbb{H}_{F_{\mathbf{b}'}, \rho}(q, t),$$

where $A \preceq B$ means that $B - A \in \mathbb{N}[q, t]$. More generally, if $x = (i, c) \leq y = (j, d)$ are cells common to both shapes, then

$$\mathbb{H}_{[x,y]_{\mathbf{b}}, \rho_{x,y}}(q, t) \preceq \mathbb{H}_{[x,y]_{\mathbf{b}'}, \rho_{x,y}}(q, t).$$

In particular, for fixed r and $b_1 \geq r$, the boundary sequences $(b_1, r, \dots, r)$ and $(b_1, \dots, b_1)$ give coefficientwise lower and upper bounds for every boundary with that top width.

Proof. The two global word sets have the same word length $b_1 - 1$. Every $\mathbf{b}$ -admissible word satisfies the defining prefix inequalities for $\mathbf{b}'$, so $\mathcal{W}_{\mathbf{b}} \subseteq \mathcal{W}_{\mathbf{b}'}$. The weights attached to a word are identical, and the first inequality follows term by term from Theorem 5.1. The local word sets have the same prescribed endpoints; the same inclusion of prefix-legal words and the local formula of Proposition 5.10 give the second inequality. $\square$

Corollary 5.15 (A universal Gaussian-binomial envelope). *Put $M = b_1 - 1$. For $1 \leq k \leq r - 1$, one has*

$$[t^k] \mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) \preceq q^{k(k-1)} \sum_{h=k}^{\min\{r-1, M-k+1\}} \begin{bmatrix} h-1 \\ k-1 \end{bmatrix}_q \begin{bmatrix} M-h+1 \\ k \end{bmatrix}_q.$$

The right-hand side is the exact t^k -coefficient for the rectangular boundary $(b_1, \dots, b_1)$. Thus every Ferrers shape with fixed top width has an explicit coefficientwise q -binomial upper envelope.

Proof. By Corollary 5.14, it suffices to evaluate the rectangular boundary $\bar{\mathbf{b}} = (b_1, \dots, b_1)$. Its admissibility inequalities are automatic: after a prefix, the column coordinate is at most $1 + \#L + \#R \leq M + 1 = b_1$. A word with h letters L and k L -runs has the decomposition

$$R^{a_0} L^{h_1} R^{a_1} \dots L^{h_k} R^{a_k},$$

where $h_i \geq 1$, $a_0, a_k \geq 0$, and $a_i \geq 1$ internally. Summing the run-start weight over the compositions gives

$$q^{k(k-1)} \begin{bmatrix} h-1 \\ k-1 \end{bmatrix}_q \begin{bmatrix} M-h+1 \\ k \end{bmatrix}_q.$$

Here $h \geq k$, $h \leq r - 1$, and $M - h \geq k - 1$, which are exactly the displayed summation bounds. Summing over h proves the claim. $\square$

Corollary 5.16 (The first t -slice for an arbitrary boundary). *For the boundary sequence $\mathbf{b}$ above, write $[m]_q = 1 + q + \dots + q^{m-1}$ for $m \geq 1$. Then*

$$[t] \mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = \sum_{i=2}^r [b_i - i + 1]_q.$$

In particular, for the trapezoidal boundary $b_i = s - i$ this becomes

$$[t] \mathbb{H}_{T_{r,s}, \rho}(q, t) = \sum_{i=2}^r [s - 2i + 1]_q.$$

Proof. A word with exactly one left run has the form $w = R^a L^h R^{b_1-1-a-h}$, where $1 \leq h \leq r - 1$ and $a \geq 0$. Its initial cell is $(h + 1, h + 1)$. The prefix inequalities before the left run are strongest at its end and read $h + 1 + a \leq b_{h+1}$; after the left run, monotonicity of the boundary makes the inequalities automatic until the final row, where $c \leq b_1$ is automatic. Thus, for $i = h + 1$, the admissible values are $0 \leq a \leq b_i - i$. The unique run start is at position $a + 1$, so its weight is q^a . Summing over $i = 2, \dots, r$ gives the first identity; the trapezoidal specialization follows from $b_i = s - i$. $\square$

Corollary 5.17 (Rigidity of the rectangular envelope). *Let $\bar{\mathbf{b}} = (b_1, \dots, b_1)$ be the rectangular boundary with the same top width as $\mathbf{b}$. Then*

$$[t] \mathbb{H}_{F_{\bar{\mathbf{b}}}, \rho}(q, t) - [t] \mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = \sum_{i=2}^r q^{b_i-i+1} [b_1 - b_i]_q.$$

Consequently, equality in the coefficientwise Gaussian-binomial envelope for the whole numerator can occur only for the rectangular boundary; in fact, equality of the first t -slice already forces $b_i = b_1$ for every $i \geq 2$.

Proof. The first-slice formula gives the difference as $\sum_{i=2}^r ([b_1 - i + 1]_q - [b_i - i + 1]_q)$. The elementary identity $[u]_q - [v]_q = q^v [u - v]_q$ for $u \geq v$ yields the displayed sum. Each summand is zero exactly when $b_i = b_1$, so equality of the first slice is equivalent to rectangularity. The final assertion follows from the envelope corollary. $\square$

Corollary 5.18 (The top t -slice for an arbitrary boundary). *Put $m = r - 1$ and, for $0 \leq j \leq m - 1$, set $B_j = b_{r-j} - r$. Then*

$$[t^{r-1}] \mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = q^{\binom{m}{2}} \sum_{\substack{0 \leq A_0 < \dots < A_{m-1} \\ A_j \leq B_j}} q^{A_0 + \dots + A_{m-1}}.$$

For $r = 1$ the displayed sum consists of the single empty sequence and equals 1. For the trapezoidal boundary $b_i = s - i$, writing $\delta = s - 2r \geq 0$ and $\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{u=1}^k (1 - q^{n-k+u}) / (1 - q^u)$, this specializes to

$$[t^{r-1}] \mathbb{H}_{T_{r,s}, \rho}(q, t) = q^{(r-1)(r-2)} \begin{bmatrix} s - r - 1 \\ r - 1 \end{bmatrix}_q.$$

Proof. Having $r - 1$ left runs forces exactly $r - 1$ left letters, each in its own run. Write a word as $R^{a_0} L R^{a_1} L \dots L R^{a_{m-1}} L R^{a_m}$, with $a_0, a_m \geq 0$ and $a_j \geq 1$ for $1 \leq j < m$. Set $A_j = a_0 + \dots + a_j$. The prefix condition at the end of the j -th right run is precisely $A_j \leq b_{r-j} - r = B_j$, and the intermediate left moves are then legal by monotonicity. The A_j are strictly increasing, and the final right run is determined and nonnegative by the endpoint width. The run starts contribute $\sum_{j=0}^{m-1} (A_j + j) = \binom{m}{2} + \sum_j A_j$, proving the first identity. In the trapezoidal case $B_j = \delta + j$. Writing $C_j = A_j - j$ gives weakly increasing integers $0 \leq C_0 \leq \dots \leq C_{m-1} \leq \delta$; their generating function is the Gaussian binomial $\begin{bmatrix} \delta + m \\ m \end{bmatrix}_q$. The outer factor contributes the second copy of $\binom{m}{2}$ in the q -exponent. $\square$

Example 5.19 (A nonlinear boundary). For $\mathbf{b} = (5, 4, 4)$, the boundary is not linear. The admissible words are

$$RRRR, LRRR, RLRR, RRLR, LLRR, LRLR, RLLR,$$

and the theorem gives

$$\mathbb{H}_{F_{\mathbf{b}}, \rho}(q, t) = 1 + (2 + 2q + q^2)t + q^2 t^2.$$

This example lies outside the linear trapezoidal subfamily and illustrates that the boundary inequalities retain genuinely shape-dependent information.

6. ADJOINING A MINIMUM

The next argument is valid for any finite graded poset with a unique maximum.

Lemma 6.1. *Let P have ranks $0, \dots, H$ and a unique maximum. Adjoin a new minimum $\widehat{0}$ and denote the resulting bounded poset by $\widehat{P}$. Give it the shifted rank*

$$\widehat{\rho}(\widehat{0}) = 0, \quad \widehat{\rho}(x) = \rho(x) + 1 \quad (x \in P).$$

Then

$$(22) \quad \mathcal{F}_{\widehat{P}, \widehat{\rho}}(q, t) = \frac{\mathcal{F}_{P, \rho}(q, qt)}{1 - t},$$

$$(23) \quad \mathbb{H}_{\widehat{P}, \widehat{\rho}}(q, t) = \mathbb{H}_{P, \rho}(q, qt).$$

Proof. A weak chain of total length m in $\widehat{P}$ has a unique initial block of $m - \ell$ copies of $\widehat{0}$, followed by a weak chain of length ℓ in P . Shifting every old rank by one multiplies the weight of that suffix by q^ℓ . Therefore

$$A_m(\widehat{P}, \widehat{\rho}; q) = \sum_{\ell=0}^m q^\ell A_\ell(P, \rho; q).$$

Summing in m gives Equation (22).

Using Equation (1), the denominator on the right-hand side of Equation (22) is

$$(1-t) \prod_{a=0}^H (1-q^a(qt)) = \prod_{j=0}^{H+1} (1-q^j t),$$

which is exactly the prescribed denominator for $\hat{\rho}$. Both sides have now been written over the same fixed denominator, so comparison of numerators proves Equation (23); no reduced-fraction assumption is required. $\square$

Corollary 6.2 (R-labelling positivity criterion). *If the minimum augmentation $\hat{P}$ of a finite graded poset P with unique maximum admits an R-labelling, then*

$$\mathbb{H}_{P,\rho}(q, t) \in \mathbb{N}[q, t].$$

Proof. Combine Proposition 6.3 with Theorem 2.3. The coefficient of a fixed monomial is a sum of nonnegative flag-descent counts, each multiplied by $q^{\sum_{s \in S} (s-1)}$. $\square$

Proposition 6.3. *With the hypotheses of Lemma 6.1,*

$$(24) \quad \mathbb{H}_{P,\rho}(q, t) = \sum_{S \subseteq [H]} \beta_{\hat{P}}(S) q^{\sigma(S) - |S|} t^{|S|}.$$

In particular, every exponent of q on the right is nonnegative.

Proof. Apply Theorem 2.2 to $\hat{P}$, then use Equation (23) and replace t by t/q . Since every $s \in S$ satisfies $s \geq 1$,

$$\sigma(S) - |S| = \sum_{s \in S} (s-1) \geq 0.$$

$\square$

7. THE AUGMENTED TYPE- B LATTICE

Let

$$\hat{L}_r = C_r \sqcup \{\hat{0}\}, \quad \hat{\rho}(\hat{0}) = 0, \quad \hat{\rho}(i, c) = c - i + 1.$$

The new element $\hat{0}$ lies below every cell.

Proposition 7.1. *The poset $\hat{L}_r$ is a bounded graded lattice. For nonbottom cells $x = (i, c)$ and $y = (j, d)$,*

$$(25) \quad x \vee y = (\min(i, j), \max(c, d)),$$

$$(26) \quad x \wedge y = \begin{cases} (\max(i, j), \min(c, d)), & \max(i, j) \leq \min(c, d), \\ \hat{0}, & \max(i, j) > \min(c, d). \end{cases}$$

Its rank function is $\hat{\rho}$.

Proof. Put $I = \min(i, j)$ and $C = \max(c, d)$. We have $I \leq C$, and

$$C \leq \max(2r - i, 2r - j) = 2r - I,$$

so the cell in Equation (25) always lies in C_r . The coordinate order in Equation (5) shows that it is the least common upper bound.

For the proposed nonbottom meet, put $J = \max(i, j)$ and $D = \min(c, d)$. When $J \leq D$,

$$D \leq \min(2r - i, 2r - j) = 2r - J,$$

so (J, D) is a cell and is the greatest common lower bound. When $J > D$, any nonbottom common lower bound would need first coordinate at least J and second coordinate at most D , contradicting the defining cell inequality that the first coordinate is at most the second. Its meet is therefore $\hat{0}$.

Among cells, a cover changes exactly one of $-i$ and c by one while remaining in C_r ; the new minimum is covered precisely by the diagonal cells (i, i) . Equivalently, if two comparable cells have rank difference at least two, changing i downward by one when possible, or increasing c by one otherwise, supplies an intermediate cell. Thus $\hat{\rho}$ is the lattice rank. $\square$

Proposition 7.2. *The lattice $\hat{L}_r$ is lower semimodular. Equivalently, its rank is supermodular:*

$$(27) \quad \hat{\rho}(x) + \hat{\rho}(y) \leq \hat{\rho}(x \wedge y) + \hat{\rho}(x \vee y)$$

for all $x, y \in \hat{L}_r$.

Proof. Pairs involving $\hat{0}$ give equality. For two cells with nonbottom meet, direct substitution of Equations (25) and (26) also gives equality.

Suppose the meet is $\hat{0}$. Interchanging x and y if necessary, assume $i \geq j$. The failed cell condition is $i > \min(c, d)$. Since $c \geq i$, it follows that $d < i \leq c$. Consequently $x \vee y = (j, c)$, and

$$\hat{\rho}(x) + \hat{\rho}(y) - \hat{\rho}(x \vee y) = d - i + 1 \leq 0.$$

Because $\hat{\rho}(x \wedge y) = 0$, this is Equation (27). $\square$

Independently of the preceding structural property, the cell coordinates give the direct labelling needed for the descent formula.

Proposition 7.3. *Label the covers of $\hat{L}_r$ by integers, with their usual order, as follows:*

$$(28) \quad \begin{aligned} \lambda(\hat{0}, (i, i)) &= i, \\ \lambda((i, c), (i-1, c)) &= -(i-1), \\ \lambda((i, c), (i, c+1)) &= c+1. \end{aligned}$$

Whenever the displayed pairs are covers, this is an EL-labelling of $\hat{L}_r$. In particular the labels along any maximal chain are pairwise distinct, so the strictly increasing chain is also the unique weakly increasing chain.

Proof. Consider first an interval from (i, c) to (j, d) , with neither endpoint equal to $\hat{0}$. Every maximal chain is a shuffle of the left-expansion labels

$$-(i-1), -(i-2), \dots, -j$$

and the right-expansion labels

$$c+1, c+2, \dots, d,$$

with each displayed list used in its given order. Both lists are strictly increasing. Because every left label is negative and every right label is positive, the unique increasing shuffle performs all left expansions before all right expansions. At every step where both moves are available, it chooses the smaller label, so it is also lexicographically first.

Now consider an interval from $\hat{0}$ to (j, d) . Its unique increasing maximal chain begins with the atom (j, j) , then makes only right expansions; its labels are

$$j, j+1, \dots, d.$$

A chain beginning at an atom (i, i) with $i > j$ must eventually make a left expansion, producing a positive-to-negative descent. Moreover, the initial label j is the smallest label of an atom below (j, d) . Hence the displayed chain is uniquely increasing and lexicographically first. Degenerate intervals are unique chains, so these cases cover every interval. $\square$

8. EL DESCENTS AND THE BALLOT-WORD FORMULA

We now identify the maximal chains and descents of the explicit labelling in Proposition 7.3. A cover after the initial atom is called an L -move if it decreases the first cell coordinate and an R -move if it increases the second.

Lemma 8.1. *Reading the L - and R -moves gives a bijection from the maximal chains of $\widehat{L}_r$ to $\mathcal{B}_r$. Under this bijection, the descent set of the label sequence is $D(w)$.*

Proof. A maximal chain begins at an atom (i, i) and ends at $(1, 2r - 1)$. After the initial edge, it has $i - 1$ left moves and $2r - 1 - i$ right moves, hence $m = 2r - 2$ moves in total. If L_k and R_k count the two moves in the first k letters, the intermediate cell is

$$(i - L_k, i + R_k).$$

Its right boundary condition is

$$(29) \quad R_k - L_k \leq 2(r - i).$$

The remaining cell inequalities $1 \leq i - L_k \leq r$ and $i - L_k \leq i + R_k$ hold automatically: there are only $i - 1$ letters L in the whole word, so the first coordinate decreases from i to 1 without leaving $\{1, \dots, r\}$. At the end equality holds in Equation (29). Consequently, Equation (29) for all prefixes is equivalent to saying that every suffix of the word has nonnegative R -minus- L balance. This is exactly the ballot-prefix condition on the reversed word.

Conversely, let $w \in \mathcal{B}_r$, and put $b = \#R - \#L$. The ballot condition gives $0 \leq b \leq 2r - 2$, and b is even. Thus $i = r - b/2$ is an integer in $[1, r]$. Starting at (i, i) and following w , the reversed ballot inequalities are exactly Equation (29); the path therefore stays in C_r and terminates at $(1, 2r - 1)$. This constructs the inverse bijection.

It remains to read the descents. The maximal chain has $2r - 1$ covers, and descents occur at internal ranks $1, \dots, 2r - 2$. The initial label is the positive integer i . Consecutive left moves have increasing negative labels, consecutive right moves have increasing positive labels, and a left move followed by a right move is also increasing. Comparing the initial label with the first move therefore produces a descent at rank 1 if and only if that move is left. For $j \geq 2$, comparing the $(j - 1)$ -st and j -th moves produces a descent at rank j if and only if a right move is followed by a left move. These are exactly the elements of $D(w)$. $\square$

Proof of Theorem 1.1. By Lemma 3.1 and Proposition 7.1, adjoining a minimum to P_r gives the bounded graded lattice $\widehat{L}_r$. The explicit EL-labelling of Proposition 7.3 is an R -labelling, so Equation (4) and Proposition 6.3 give

$$\mathbb{H}_{P_r, \text{rk}}(q, t) = \sum_{\mathfrak{c}} q^{\sigma(\text{Des}(\mathfrak{c})) - |\text{Des}(\mathfrak{c})|} t^{|\text{Des}(\mathfrak{c})|},$$

where the sum is over the maximal chains of $\widehat{L}_r$. Apply Lemma 8.1: the chains become the words $w \in \mathcal{B}_r$, their descent sets become $D(w)$, and

$$\sigma(D(w)) - |D(w)| = \sum_{j \in D(w)} (j - 1).$$

This is the asserted formula. Every summand is a monomial with coefficient one and nonnegative exponents, proving coefficientwise nonnegativity as well. For $r = 1$, the unique chain corresponds to the empty word and the formula gives $\mathbb{H}_{P_1, \text{rk}}(q, t) = 1$. $\square$

Corollary 8.2 (Type- C companion formula). *For the standard positive-root poset $P_r^C = \Phi^+(C_r)$, graded by root height minus one and with Chapoton's denominator convention, one has*

$$\mathbb{H}_{P_r^C, \text{rk}}(q, t) = \sum_{w \in \mathcal{B}_r} q^{\sum_{j \in D(w)} (j-1)} t^{|D(w)|}.$$

In particular, its q -Zeta numerator is coefficientwise nonnegative for every $r \geq 1$.

Proof. By Lemma 3.2, the ranked posets P_r^C and P_r are isomorphic. The q -Zeta numerator is defined from rank-weighted weak-chain enumeration, hence is invariant under a rank-preserving poset isomorphism. The assertion follows from Theorem 1.1. $\square$

The formula gives sharp global information that is not visible from nonnegativity alone.

Corollary 8.3. *For every $r \geq 1$,*

$$\mathbb{H}_{P_r, \text{rk}}(1, 1) = \binom{2r-2}{r-1}, \quad \deg_t \mathbb{H}_{P_r, \text{rk}} = r-1,$$

and the coefficient of the highest power of t is

$$[t^{r-1}] \mathbb{H}_{P_r, \text{rk}}(q, t) = q^{(r-1)(r-2)}.$$

Moreover, for $0 \leq k \leq r-1$,

$$[t^k] \mathbb{H}_{P_r, \text{rk}}(1, t) = \binom{r-1}{k}^2.$$

8.1. Spectral consequences at $q = 1$. The preceding specialization has useful one-variable and asymptotic consequences.

Corollary 8.4 (Real-rootedness at $q = 1$). *For $r \geq 2$, the polynomial*

$$\mathbb{H}_{P_r, \text{rk}}(1, t) = \sum_{k=0}^{r-1} \binom{r-1}{k}^2 t^k$$

has $r-1$ distinct negative real zeros. In particular, its coefficient sequence is strictly log-concave in the interior and unimodal.

Proof. Put $n = r-1$. The classical Legendre identity gives

$$\sum_{k=0}^n \binom{n}{k}^2 t^k = (1-t)^n P_n\left(\frac{1+t}{1-t}\right),$$

where P_n is the Legendre polynomial. The n zeros of P_n are simple and lie in $(-1, 1)$. Under the inverse fractional-linear change $t = (x-1)/(x+1)$, this interval maps bijectively to the negative real axis, and the displayed identity shows that no additional finite zeros occur. Strict log-concavity and unimodality follow from Newton's inequalities. $\square$

Corollary 8.5 (γ -positivity at $q = 1$). *Let $n = r-1$. Then*

$$\mathbb{H}_{P_r, \text{rk}}(1, t) = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \binom{2j}{j} t^j (1+t)^{n-2j}.$$

In particular, the $q = 1$ specialization is γ -positive, with $\gamma_j = \binom{n}{2j} \binom{2j}{j}$.

Proof. By Corollary 8.3, it remains to rewrite $\sum_{k=0}^n \binom{n}{k}^2 t^k$. The coefficient of t^k in the right-hand side is

$$\sum_{j=0}^{\min\{\lfloor n/2 \rfloor, k, n-k\}} \frac{n!}{j!^2 (k-j)! (n-k-j)!} = \binom{n}{k} \sum_j \binom{k}{j} \binom{n-k}{j} = \binom{n}{k}^2,$$

where the last equality is Vandermonde's identity. This proves the displayed expansion and the asserted γ -positivity. $\square$

Corollary 8.6 (Sharp q -degree in each t -slice). *Put $n = r - 1$. For $0 \leq k \leq n$, the t^k -coefficient satisfies*

$$[t^k]\mathbb{H}_{P_r, \text{rk}}(q, t) \in q^{k(k-1)}\mathbb{N}[q], \quad \deg_q[t^k]\mathbb{H}_{P_r, \text{rk}}(q, t) = k(2n - k - 1).$$

Moreover, the coefficient of the highest power $q^{k(2n-k-1)}$ is equal to 1.

Proof. Write the elements of $D(w)$ as $j_1 < \dots < j_k$. Two consecutive left-run starts are separated by at least one right letter, so $j_a \geq 2a - 1$. Hence

$$\sum_{a=1}^k (j_a - 1) \geq \sum_{a=1}^k (2a - 2) = k(k - 1).$$

For the upper bound, reverse w and write $m = 2n$. A marked position in $D(w)$ becomes either a valley, an L followed by an R , or a terminal L in the ballot word. If these marked positions are $p_1 < \dots < p_k$, the ballot condition and the initial R imply $p_a \geq 2a$. Therefore

$$\sum_{a=1}^k (j_a - 1) = \sum_{a=1}^k (m - p_a) \leq \sum_{a=1}^k (m - 2a) = k(2n - k - 1).$$

Equality forces the reversed word to be

$$(RL)^k R^{2n-2k},$$

whose reversal is in $\mathcal{B}_r$ and has the required run starts. Thus the upper degree is attained by exactly one word, proving the final assertion. $\square$

Lemma 8.7. *Let $a(n, k)$ be the number of words of length $2n$ whose every prefix has at least as many R 's as L 's and which have exactly k L -runs. Then*

$$a(n, k) = \binom{n}{k}^2.$$

Proof. Let $C(z, u)$ be the generating function of Dyck words, with z marking semilength and u marking L -runs, using the same convention that R is an up-step. The first-return decomposition gives

$$C = 1 + zC(C + u - 1).$$

Indeed, in a first-return factor $RALB$, the outer pair creates a new L -run exactly when A is empty. (The resulting generating function is the classical Narayana generating function for Dyck paths by number of peaks; see [13].) Every ballot word has a unique factorization

$$D_0 R D_1 R \cdots R D_h,$$

where the D_i are Dyck words and h is the final height. Taking the even-length part (so $h = 2j$) therefore gives the generating function

$$E(z, u) = \sum_{n, k} a(n, k) z^n u^k = \frac{C(z, u)}{1 - zC(z, u)^2}.$$

Put $Y = C - 1$, so $Y = z(1 + Y)(u + Y)$. The coefficients of E are extracted from this functional equation by Lagrange–Bürmann inversion. First expand

$$E(z, u) = \sum_{j \geq 0} z^j C(z, u)^{2j+1}.$$

For $s \geq 1$, $p \geq 1$, and $1 \leq k \leq s$, inversion applied to $Y = z(1 + Y)(u + Y)$ gives

$$\begin{aligned} [z^s u^k] C(z, u)^p &= \frac{p}{s} [y^{s-1} u^k] (1 + y)^{p+s-1} (u + y)^s \\ &= \frac{p}{s} \binom{s}{k} \binom{p+s-1}{k-1}. \end{aligned}$$

Consequently, for $n \geq 1$ and $1 \leq k \leq n$, with $s = n - j$,

$$[z^n u^k]E = \sum_{s=k}^n \frac{2n - 2s + 1}{s} \binom{s}{k} \binom{2n - s}{k - 1}.$$

To evaluate this finite sum, use $\binom{s}{k}/s = \binom{s-1}{k-1}/k$, put $A = s - 1$ and $B = 2n - 1 - A$, and apply Pascal's identity:

$$\begin{aligned} & \frac{B - A}{k} \binom{A}{k-1} \binom{B}{k-1} \\ &= \binom{A}{k-1} \binom{B}{k} - \binom{A}{k} \binom{B}{k-1} \\ &= \binom{A+1}{k} \binom{2n - A - 1}{k} - \binom{A}{k} \binom{2n - A}{k}. \end{aligned}$$

The sum therefore telescopes from $A = k - 1$ to $A = n - 1$, and is $\binom{n}{k}^2$. For $k = 0$, setting $u = 0$ gives $C(z, 0) = 1$ and $E(z, 0) = 1/(1 - z)$, so the coefficient is 1; for $n = 0$ the empty word gives the same conclusion. Hence $[z^n u^k]E(z, u) = \binom{n}{k}^2$ in all cases. $\square$

Proof of Corollary 8.3. Write $n = r - 1$. Reversal identifies $\mathcal{B}_r$ with the ballot words of length $2n$. By the reflection principle, the number ending at height $2k$ is

$$\binom{2n}{n-k} - \binom{2n}{n-k-1} \quad (0 \leq k \leq n),$$

where $\binom{2n}{-1} = 0$. Summing over k telescopes to $\binom{2n}{n}$, proving the first identity.

Every reversed ballot word has at most n letters L , so $|D(w)| \leq n$. Equality forces exactly n letters L , each starting a separate run. For $n = 0$ the unique word is empty. For $n \geq 1$, a nonempty word in $\mathcal{B}_r$ ends in R , and the unique such word with n isolated letters L is

$$w = (LR)^n.$$

Its descent positions are $1, 3, \dots, 2n - 1$, and hence its q -exponent is

$$\sum_{a=0}^{n-1} 2a = n(n-1).$$

This proves both the sharp degree and the stated leading coefficient.

It remains to evaluate the coefficients at $q = 1$. By Theorem 1.1, the coefficient of t^k is the number of ballot words of length $2n$ with exactly k runs of the letter L ; reversal preserves the number of runs. Lemma 8.7 therefore yields

$$[t^k]\mathbb{H}_{P_r, \text{rk}}(1, t) = \binom{r-1}{k}^2.$$

$\square$

Corollary 8.8 (Interlacing across Lie rank). *Set*

$$B_n(t) = \sum_{k=0}^n \binom{n}{k}^2 t^k \quad (n \geq 0).$$

For every $n \geq 1$, the n zeros of B_n and the $n+1$ zeros of B_{n+1} are all simple, negative, and strictly interlace on $(-\infty, 0)$. Equivalently, the $q = 1$ specializations of the type- B/C numerators form a Sturm sequence after the rank shift $n = r - 1$.

Proof. The Legendre identity used in Corollary 8.4 gives

$$B_n(t) = (1-t)^n P_n\left(\frac{1+t}{1-t}\right).$$

The fractional-linear map $t \mapsto (1+t)/(1-t)$ is strictly increasing from $(-\infty, 0)$ onto $(-1, 1)$. Consecutive Legendre polynomials have simple zeros which strictly interlace in $(-1, 1)$, by the Sturm separation theorem for their differential equation. Transporting these zeros through the increasing map proves the claim. $\square$

Corollary 8.9 (Rank generating function and recurrence). *With $B_n(t)$ as in Corollary 8.8, one has*

$$\sum_{n \geq 0} B_n(t) z^n = \frac{1}{\sqrt{1 - 2(1+t)z + (1-t)^2 z^2}}.$$

Equivalently, $B_0(t) = 1$, $B_1(t) = 1 + t$, and, for $n \geq 1$,

$$(n+1)B_{n+1}(t) = (2n+1)(1+t)B_n(t) - n(1-t)^2 B_{n-1}(t).$$

Proof. The Legendre generating function

$$\sum_{n \geq 0} P_n(x) u^n = (1 - 2xu + u^2)^{-1/2}$$

and the identity in Corollary 8.8, with $x = (1+t)/(1-t)$ and $u = (1-t)z$, give the displayed generating function. The recurrence follows in the same way from

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x),$$

after multiplying by $(1-t)^{n+1}$. $\square$

Corollary 8.10 (Asymptotic zero distribution). *Let $\tau_{n,1}, \dots, \tau_{n,n}$ be the zeros of $B_n(t)$, counted without repetition. For every continuous compactly supported function f on $(-\infty, 0)$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n f(\tau_{n,j}) = \frac{1}{\pi} \int_{-\infty}^0 \frac{f(t)}{\sqrt{-t}(1-t)} dt.$$

Thus the rank-shifted type-B/C zero measures have the explicit push-forward of the arcsine law as their limiting density. If the zeros are ordered increasingly and $j_n/n \rightarrow u \in (0, 1)$, then

$$\tau_{n,j_n} \longrightarrow -\cot^2\left(\frac{\pi u}{2}\right).$$

Proof. The normalized zero-counting measures of the Legendre polynomials P_n converge to the arcsine measure

$$\frac{dx}{\pi\sqrt{1-x^2}}, \quad -1 < x < 1,$$

see [14]. By the identity in Corollary 8.8, the zeros of B_n are the images of the zeros of P_n under

$$t = \frac{x-1}{x+1}.$$

The change of variables $x = (1+t)/(1-t)$ gives

$$\frac{dx}{\pi\sqrt{1-x^2}} = \frac{dt}{\pi\sqrt{-t}(1-t)}$$

on $(-\infty, 0)$. Applying the zero-distribution convergence to the pull-back of f proves the formula. The limiting distribution function is

$$F(t) = \frac{2}{\pi} \arctan \frac{1}{\sqrt{-t}}, \quad t < 0,$$

which is continuous and strictly increasing. Quantile convergence for the ordered zeros therefore gives $F^{-1}(u) = -\cot^2(\pi u/2)$. $\square$

For completeness, the flag interpretation also yields a three-rank topological refinement. Recall that $H = 2r - 2$. For $0 \leq a < b < c \leq H - 1$, let L_a be rank a of P_r , set $f_a = |L_a|$, let e_{ab} count comparable pairs in $L_a \times L_b$, and let n_{abc} count chains $x < y < z$ through the three ranks.

Corollary 8.11. *Define*

$$\beta_{abc} = n_{abc} - e_{ab} - e_{ac} - e_{bc} + f_a + f_b + f_c - 1.$$

Then

$$[t^3]\mathbb{H}_{P_r, \text{rk}}(q, t) = \sum_{0 \leq a < b < c \leq H-1} \beta_{abc} q^{a+b+c}, \quad \beta_{abc} \geq 0.$$

Moreover, β_{abc} is the reduced Euler characteristic of the order complex induced on $L_a \cup L_b \cup L_c$.

Proof. Apply Boolean inversion in Equation (2) to the set $\{a+1, b+1, c+1\}$, then use Equation (24). The induced three-rank order complex has $f_a + f_b + f_c$ vertices, $e_{ab} + e_{ac} + e_{bc}$ edges, and n_{abc} triangles, giving the Euler-characteristic formula. Nonnegativity follows from the descent interpretation in Equation (4). The Euler interpretation alone would not determine the sign. $\square$

9. FURTHER REMARKS

9.1. Position in the literature. The order-theoretic structure of positive roots, including their Hasse diagrams and antichain symmetries, has been studied extensively; a useful general reference is Panyushev [11]. Generalized Catalan enumeration for Weyl groups and root-poset antichains is developed in [1]. The statistic considered here is different: it weights weak chains by the sum of their root-poset ranks.

Chapoton introduced this q-Zeta invariant, established the bounded flag expansion, and observed that the type- B numerator appeared to have nonnegative coefficients which, as polynomials in q , specialise at $q = 1$ to squares of binomial coefficients [3]. Under the Lie-rank convention of this paper, the proved specialization is $[t^k]\mathbb{H}_{P_r, \text{rk}}(1, t) = \binom{r-1}{k}^2$. The descent formula used above goes back to Stanley's work on Jordan–Hölder sets [12]. Björner's EL-labelling framework packages intervalwise increasing chains together with lexicographic shellings [2]. In the present setting, the signed endpoint labelling makes the relevant descent enumerator completely explicit and leads to the ballot-word formula in Theorem 1.1. Under the rank and denominator conventions fixed above, this gives an all-rank proof of the type- B positivity statement highlighted in Chapoton's discussion.

Recent work relates classical zeta polynomials of lattice-point posets to Ehrhart polynomials of preorder polytopes [4]. That unrefined Ehrhart–zeta duality is complementary to our rank-sensitive q-Zeta positivity theorem and its explicit Ferrers-cell EL/ballot model.

The ballot encoding also places the result next to the established theory of q -ballot polynomials, where boundary statistics on lattice paths provide natural q -analogues of Catalan and Narayana enumerators [7]. Our statistic is different: it is the descent weight forced by Chapoton's rank-sensitive chain enumerator, and the trapezoidal cell model explains why this weight remains nonnegative in every rank. Classical two-boundary path formulas by Krattenthaler and Mohanty also count descents and major index [10]; the Ferrers theorem here uses the same path language but derives the q -weight from the poset's fixed denominator normalization. The resulting refinement is also adjacent to the descent-refined major-index enumeration of two-row standard Young tableaux, for which closed forms and unimodality phenomena are known [9]. The present ballot model is not an immediate specialization of those tableau formulas: its weight is the reverse-major statistic dictated by the q-Zeta denominator. This distinction identifies a concrete bridge for future q-hypergeometric refinements while keeping the current theorem self-contained.

The two-parameter theorem in Theorem 4.1 isolates the part of the argument that is genuinely order-theoretic. Widening the right boundary from $2r - i$ to $s - i$ preserves the lattice and EL structures, while the ballot condition records exactly which starting atoms remain available. Thus the type- B/C root-poset result is the balanced member of a larger positive family, providing a reusable test class for the positivity question in Chapoton's q -Zeta theory rather than a one-off root-system computation. The new slice-degree corollary shows that this extension retains sharp coefficient-level control: the extremal q -weight depends explicitly on the width parameter s , while the leading monomial remains rigid. The same family admits a closed specialization $\mathbb{H}_{T_{r,s},\rho}(1,1) = \binom{s-2}{r-1}$, which quantifies the total chain contribution as the boundary widens. The endpoint decomposition in Theorem 4.2 complements the two extreme slice formulas: after fixing the numbers of left and right steps, the reversed-ballot condition becomes a single linear boundary, and the q -Zeta weight is exactly the ordinary major index of the move word. Krattenthaler–Mohanty's one-boundary formula therefore supplies a closed Gaussian-binomial expression for every intermediate slice, while the initial-left-run correction accounts for the augmented atom. This makes the bridge to classical q -enumeration precise without identifying the q -Zeta statistic with a standard q -Narayana statistic. The interval theorem Theorem 4.3 shows that this closed form is genuinely local as well: once the lower endpoint is nonbottom, all dependence on the ambient trapezoid is compressed into the displacement and the single boundary slack δ . At $q = 1$, the numerator at $t = 1$ is the reflection-principle difference $\binom{\ell+u}{\ell} - \binom{\ell+u}{\ell+\delta+1}$, so the local q -Zeta invariant interpolates between a Gaussian major-index polynomial and a classical ballot count. Together with Theorem 4.5, this gives a closed formula for every interval of the augmented trapezoidal lattice, including those whose lower endpoint is the adjoined minimum. The latter are finite sums over admissible starting atoms, so no additional boundary case is hidden in the interval statement.

The $q = 1$ profile admits a complementary generating-function description. The rational kernel in Proposition 4.9 places the family beside the generalized Narayana polynomials whose real-rootedness and interlacing were studied by Chen, Yang, and Zhang [5]. The identification is not literal: their subtracted two-parameter Narayana polynomial is a different statistic, whereas our $J_{A,B}$ arises as the $q = 1$ specialization of a rank-sensitive q -Zeta numerator and carries a Gaussian-binomial q -lift before specialization. The rational kernel therefore supplies a precise bridge to that literature without conflating the two enumerators.

The same kernel has a useful fixed-offset consequence that is not visible from a single trapezoid. Constant-term extraction gives the algebraic series in Proposition 4.10; in particular, every fixed offset $J_{n,n+\kappa}(t)$ has the common exponential rate $(1 + \sqrt{t})^2$ for $t \geq 0$. Thus the finite-rank Jacobi picture and the two-parameter recurrence are compatible with an exact algebraic growth law throughout the fixed-offset regimes, providing a direct entry point for further asymptotic questions.

The Ferrers-cell theorem Theorem 5.1 identifies the actual universality class behind these examples. The boundary need only be nonincreasing; no linearity or root-system input is required. The q -Zeta numerator is then a positive descent enumerator over the admissible cell paths, so the root-poset and trapezoidal formulas are specializations of one boundary-stable mechanism. In particular, the standard type- A_r root poset is the constant-boundary member $F_{(r,\dots,r)}$, so the same mechanism covers all three classical families A , B , and C in one coordinate model. Moreover, the same labelling restricts to every interval, giving the interval-stable positivity statement in Corollary 5.9; positivity is therefore a local property throughout this lattice family, not only a global feature of its top interval. Thus the theorem gives an explicit infinite family of bounded graded, non-distributive lattices ($r \geq 3$) realizing Chapoton's R -labelling positivity criterion intervalwise. The shape-monotonicity corollary adds a comparison principle within this universality class. At fixed top width, moving the lower boundary outward does not alter the weight of any existing path; it only admits additional paths. The resulting coefficientwise order is therefore stronger than a comparison after specializing q or t . The first-slice identity is a complementary sharpness statement: regardless of the later-run

interactions, all one-run contributions separate by their starting row and sum to q -integers determined by the boundary. At $q = 1$, the type- B/C Narayana specialization is not only real-rooted but has the explicit γ -coefficients $\binom{r-1}{2j} \binom{2j}{j}$, making its symmetry and unimodality transparent in the standard γ -basis. The top-slice formula is the dual extreme in the run decomposition: all left runs are forced to have length one, while the allowed cumulative right-run positions form a flagged strict sequence. For the trapezoidal boundary this flag becomes rectangular after subtracting the index, explaining the Gaussian binomial specialization. The interval version shows that this is not an artifact of the top element: for every nonbottom interval, the coefficient of the largest possible power of t is controlled by the same flagged strict-sequence mechanism, with the endpoint displacement supplying the additional truncation. The envelope corollary complements this local statement globally: for every fixed descent number, the rectangular boundary gives a finite Gaussian-binomial convolution that bounds all lower Ferrers shapes coefficientwise.

The $q = 0$ edge also has a classical order-theoretic shadow. Applying Chapoton's characteristic-polynomial identity to the augmented lattice and using the initial-descent part of the ballot model gives $X_{\widehat{F}_b}(y) = y^{b_1} - ry^{b_1-1} + (r-1)y^{b_1-2}$. Thus all intermediate rows disappear from this Möbius invariant, even though they affect the positive q -Zeta numerator. This separates the boundary data seen by the rank-sensitive refinement from the much coarser characteristic polynomial and provides a useful consistency check on the EL description. Characteristic polynomials also occur in the arrangement theory of ideals of classical root systems [15]; that literature concerns hyperplane and toric arrangements, whereas $X_{\widehat{F}_b}$ is the Möbius polynomial of the augmented root-poset lattice itself. The two specializations should therefore not be conflated.

9.2. Finite verification. As an independent exact-arithmetic check, we generated the three families of positive roots of B_r in simple-root coordinates, matched them bijectively to C_r , and verified the coordinate order, the join and meet formulae, and the lower-semimodular rank inequality for $1 \leq r \leq 10$. The same check enumerates every interval of $\widehat{L}_r$ to verify the EL property, computes the q -Zeta numerator directly from the defining weak-chain sums, and matches it against the ballot-word formula, again for $1 \leq r \leq 10$. For these ranks the word counts agree with $\binom{2r-2}{r-1}$, the sharp top t -coefficient agrees with Corollary 8.3, and the coefficients of t^k at $q = 1$ agree with $\binom{r-1}{k}^2$. The small-rank polynomials displayed in Section 1 were checked separately.

These computations are exact finite reproduction for the named ranks only. They neither prove Theorem 1.1 nor justify extrapolation to higher rank. The all-rank formula rests on the proofs in Sections 6 to 8.

9.3. Scope. The root-system result concerns the full positive-root posets of standard Lie types B_r and C_r , with r equal to the Lie rank, and Chapoton's fixed denominator normalization. The paper also proves the separate two-parameter trapezoidal-cell theorem Theorem 4.1. The type- C statement uses the known B/C root-poset isomorphism [8], made explicit here as a rank-preserving cell-poset map; it is not inferred from a finite check or a heuristic analogy. The argument neither characterizes all posets with nonnegative q -Zeta numerator nor settles the corresponding questions for type D , exceptional Coxeter types, or other denominator conventions.

10. CONCLUSION

The q -Zeta numerators of the type- A_r , type- B_r , and type- C_r positive-root posets are instances of one Ferrers-cell descent enumerator. An exact minimum-adjunction identity and a signed EL-labelling prove coefficientwise positivity in every Lie rank, while also giving the $q = 1$ Narayana-square specialization, sharp t -degrees, and unique leading monomials. The two-parameter trapezoidal theorem extends the mechanism beyond the balanced root-poset boundary and yields the total specialization $\binom{s-2}{r-1}$ together with the extremal degree $k(s - k - 3)$.

The Ferrers theorem isolates the universal order-theoretic content: arbitrary nonincreasing boundaries, all bounded intervals, and coefficientwise shape monotonicity are covered by the same ballot-word model. The first slice is the row statistic $\sum_{i=2}^r [b_i - i + 1]_q$, the highest slice is a flagged strict-sequence enumerator (Gaussian in the trapezoidal case), and every intermediate slice obeys the rectangular Gaussian-binomial envelope. The two-parameter trapezoidal family also has a finite Gaussian-binomial endpoint decomposition for every intermediate slice, with an alternating q -binomial formula at $t = 1$. Every nonbottom interval of the trapezoidal family has the same one-boundary Gaussian formula after replacing the global endpoint data by its displacement and boundary slack; its $q = 1$ value is an explicit ballot-number difference. Principal intervals from the adjoined minimum are finite sums of the same Gaussian contributions, so all intervals in the augmented trapezoidal lattice are covered by closed formulas. At $q = 1$ the type- B/C specialization is γ -positive, while the type- A companion has the parallel negative-real-root consequence stated in Corollary 5.6; its rank sequence also satisfies a second-order recurrence and has strictly interlacing negative zeros (Corollary 5.7), and its normalized zero measures converge to the same explicit density. In addition, the Narayana-square specializations strictly interlace from one Lie rank to the next, providing a rank-to-rank Sturm structure that is invisible in coefficientwise positivity alone. The associated rank generating function is algebraic and yields an explicit three-term recurrence, making the $q = 1$ specialization directly accessible to further asymptotic and spectral analysis. The zero-counting measures also have a closed rank-asymptotic limit, obtained by pushing the Legendre arcsine law through the same Cayley transform. As an additional classical specialization, the augmented Ferrers lattice has characteristic polynomial $y^{b_1} - ry^{b_1-1} + (r-1)y^{b_1-2}$; its independence from the lower boundary contrasts with the boundary-sensitive positive q -Zeta expansion.

ACKNOWLEDGEMENTS

We acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

DISCLOSURE OF AUTOMATED ASSISTANCE

OpenAI GPT-5.6 Sol, Anthropic Claude Fable 5, and Grok 4.6 were used in a generative-AI workflow to assist with the mathematical development and preparation of this manuscript. The assistance included formulating the Ferrers-cell model for nondecreasing boundary sequences; deriving intermediate steps for the signed EL-labellings and the reversed-ballot-word formula for Chapoton's q -Zeta numerators; working out Gaussian-binomial endpoint and interval decompositions for trapezoidal and general Ferrers families; developing the transfer-matrix recursion and the $q = 1$ Jacobi-Legendre, zero-interlacing, generating-function, and asymptotic consequences; constructing low-rank examples; organizing the theorem and proof dependencies across Sections 2–8; and drafting and revising the abstract, exposition, LaTeX source, and bibliography. The results await further verification by domain experts.

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